A Project Report on

AI Powered Lead Generation & Sales Automation Platform

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

By

Durvesh Kanade: 22106013

Aaryaman Kattali: 22106077

Vaishnavi Dumbre: 22106115

Ritik Pandey: 22106054

Under the Guidance of

Prof. Viki Patil

Department of Computer Science & Engineering

(Artificial Intelligence & Machine Learning)

A.P. SHAH INSTITUTE OF TECHNOLOGY, THANE

UNIVERSITY OF MUMBAI

2025-2026

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

CERTIFICATE

This is to certify that the project entitled "**AI Powered Lead Generation & Sales Automation Platform**" is a bonafide work of **Durvesh Kanade (22106013), Aaryaman Kattali (22106077), Vaishnavi Dumbre (22106115), Ritik Pandey (22106054)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning)**.

Prof. Viki Patil Guide	Prof. Sayali P. Badhan Project Co-ordinator
Dr. Jaya Gupta Head of Department	Dr. Uttam D. Kolekar Principal

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

Project Report Approval for B.E.

This project report entitled ***AI Powered Lead Generation & Sales Automation Platform*** by ***Durvesh Kanade (22106013), Aaryaman Kattali (22106077), Vaishnavi Dumbre (22106115), Ritik Pandey (22106054)*** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science & Engineering (Artificial Intelligence & Machine Learning)***, ***2025-26***.

Examiner Name	Signature
1._____	_____
2._____	_____

Date:

Place:

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Durvesh Kanade (22106013)

Aaryaman Kattali (22106077)

Vaishnavi Dumbre (22106115)

Ritik Pandey (22106054)

Date:

Abstract

This project report presents an AI-powered lead generation and sales automation platform designed for businesses offering Salesforce implementation services in the real estate sector. The system aims to automate up to 90% of the lead identification, qualification, and outreach processes. It uses Artificial Intelligence (AI) and Natural Language Processing (NLP) to analyse user comments and detect sentiments such as happiness, anger, or dissatisfaction, enabling deeper customer insights and better decision-making. Salesforce Agentforce, an embedded conversational AI bot, manages real-time chat interactions, qualifies leads, and schedules meetings. All lead and engagement data is automatically pushed to Salesforce CRM, creating a centralized and reliable data source.

In today's competitive digital environment, traditional lead generation methods often struggle to scale due to their reliance on static websites and manual outreach. This platform overcomes these limitations by integrating machine learning-based lead scoring with automated engagement workflows, allowing businesses to identify high-potential prospects more effectively. The system analyzes firmographic and behavioral data, including user interactions and engagement patterns, using both rule-based and predictive models to improve lead qualification. By continuously evaluating these data points, the platform optimizes marketing strategies, reduces manual effort, and accelerates the sales cycle, resulting in higher-quality leads and improved conversion rates.

Additionally, the platform includes continuous learning mechanisms that adapt to changing market conditions and customer behavior. Through feedback from user interactions and sales outcomes, the system refines its predictive models and outreach strategies to enhance accuracy and engagement. The integration of AI with CRM systems improves operational efficiency and supports data-driven decision-making across the entire sales funnel, enabling organizations to reduce costs, respond faster to opportunities, and maintain a competitive advantage in dynamic digital markets.

Keywords: Sales Automation, Lead Generation, Salesforce, Agentforce, CRM (Customer Relationship Management), Digital Transformation, Machine Learning.

CONTENTS

1. Introduction 1
1.1 Background of the System 2
1.2 System Overview and Key Components 2
1.3 Motivation 3
2. Literature Survey 5
2.1 Evolution of CRM and AI Integration 5
2.2 Marketing Automation and Predictive Analytics 6
2.3 Social Media Analytics, Data Thinking, and Ethical AI 6
2.4 Salesforce Capabilities, Automation, and Business Impact 7
2.5 Conversational AI, Machine Learning Models, and Interpretability 7
2.6 Infrastructure, Advanced Capabilities, and Future of AI-Driven CRM 8
3. Limitation of Existing System 15
3.1 Data Dependency, Privacy, and Accuracy Constraints 15
3.2 Technical, Automation, and Integration Challenges 16
3.3 Scalability, Maintenance, and Performance Limitations 16
4. Problem Statement and Objective 18
4.1 Problem Statement 18
4.2. Objectives 19
4.2.1 Automation and Optimization of Lead Generation 19
4.2.2 Development of a Scalable and Intelligent Sales Pipeline 19
5. Proposed System 20
5.1 Analysis / Framework/ Algorithm 20
5.2 Modules of Proposed System 22
5.2.1. Data Acquisition Module 22
5.2.2. AI Lead Scoring Module 22
5.2.3. Content Personalization Module 23
5.2.4. News Intelligence Module 23
5.2.5. AI Chatbot Module 24
5.3 Algorithm of Proposed System 24
5.4 Workflow of the Proposed System 28
5.5 Advantages of the Proposed System 29
6. Experimental Setup 31

6.1 Overview 31
6.2 Details of Database and Input Data 32
6.3 Performance Evaluation Parameters 33
6.3.1 Machine Learning Metrics 33
6.3.2 System Performance Metrics 33
6.4 Software Setup 34
6.5 Hardware Setup 36
7. Results and Discussion 37
7.1 Performance Evaluation Metrics 43
1. Precision 43
2. Recall 43
3. F1-Score 44
4. Accuracy 44
5. Response Latency 44
7.2 Evaluation Strategy 44
8. Conclusion and Future Work 46
8.1 Future Work 47
References 48
Appendix 50
Publication 54
References 68
Acknowledgements 69

LIST OF FIGURES

Figure 5.1 Proposed Modular System Architecture……...21
Figure 5.2 Lead Classification Time-Series Graph……...23
Figure 5.3 Articles in News Intelligence Module…….24
Figure 5.4 Hybrid Lead Scoring Classification Pipeline……..26
Figure 5.5 Level 1 Data Flow Diagram (DFD)…….28
Figure 5.6 Intent Recognition and System Workflow……...29
Figure 6.1 Dataset Analysis and Label Distribution…….33
Figure 6.2 GSBG Lead Generation Technology Stack……..35
Figure 7.1 GSBG Platform Homepage Interface……...39
Figure 7.2 Salesforce Consulting Services Navigation Menu……...39
Figure 7.3 Salesforce Cloud Products Dropdown Menu……...40
Figure 7.4 Automated Content Personalization UI……...40
Figure 7.5 Agentforce AI Chatbot Welcome Screen……41
Figure 7.6 Chatbot Lead Qualification and Interaction……41
Figure 7.7 Contact Form for Manual Lead Submission……42
Figure 7.8 User Data Entry for Service Inquiry……42
Figure 7.9 Inquiry Submission Confirmation Screen……42
Figure 7.10 Automated Follow-up Email via SMTP Integration……43

LIST OF TABLES

Table 3.1 Comparison of Existing Methodologies with Proposed Framework……...17

Table 6.1 Dataset Distribution……..…31

Table 6.2 Optimized Hyperparameters for Machine Learning Classifier…….....32

Table 7.1 Comparison of Our Model with Traditional Lead Management Approaches…...45

LIST OF EQUATIONS

Equation 1 Cosine Similarity for Context Retrieval……..24
Equation 2 Hybrid Lead Score Calculation……...26
Equation 3 Precision Metric Calculation……...43
Equation 4 Recall Metric Calculation……43
Equation 5 F1-Score Metric Calculation……...44
Equation 6 Accuracy Metric Calculation……...44

ABBREVATION

AI	*Artificial Intelligence*
API	*Application Programming Interface*
AWS	*Amazon Web Services*
B2B	*Business-to-Business*
CRM	*Customer Relationship Management*
CSV	*Comma-Separated Values*
DFD	*Data Flow Diagram*
FN	*False Negative*
FP	*False Positive*
GB	*Gigabyte*
GCP	*Google Cloud Platform*
GDPR	*General Data Protection Regulation*
GPT	*Generative Pre-trained Transformer*
JSON	*JavaScript Object Notation*
KPI	*Key Performance Indicator*
LLM	*Large Language Model*
ML	*Machine Learning*
NLP	*Natural Language Processing*
NoSQL	*Not Only SQL*

RAG	*Retrieval-Augmented Generation*
SFA	*Sales Force Automation*
SHAP	*Shapley Additive exPlanations*
SMTP	*Simple Mail Transfer Protocol*
SQL	*Structured Query Language*
SSD	*Solid State Drive*
TN	*True Negative*
TP	*True Positive*
UI	*User Interface*

Chapter 1

Introduction

The GSBG AI-Powered Lead Generation & Sales Automation Platform is an advanced, end-to-end solution designed to revolutionize client acquisition for organizations delivering Salesforce implementation services, particularly within the real estate sector. In an increasingly competitive and digitally driven marketplace, businesses are required to identify high-quality leads faster, engage them more effectively, and convert them with minimal delay. This platform is specifically engineered to address these needs by automating up to 90% of the lead identification, qualification, and outreach processes, thereby enabling sales teams to concentrate on strategic, high-value interactions rather than repetitive operational tasks.

At its core, the platform integrates artificial intelligence, data analytics, and automation technologies with Salesforce's ecosystem, including Agentforce, to create a seamless and intelligent sales pipeline. The system not only identifies potential prospects but also evaluates their relevance, predicts their intent, and initiates personalized communication at scale. By transforming traditional sales workflows into a data-driven and automated process, the platform significantly enhances efficiency, accuracy, and scalability.

Furthermore, the platform ensures that all lead interactions, behavioral insights, and engagement data are captured and synchronized in real-time within Salesforce. This establishes a single source of truth, enabling organizations to make informed decisions, track performance, and continuously refine their sales strategies. The result is a modern, intelligent, and adaptive sales ecosystem capable of accelerating conversion rates and improving overall business outcomes.

1.1 Background of the System

The evolution of digital technologies and the rapid adoption of cloud-based CRM systems have fundamentally reshaped how businesses approach sales and customer relationship management. Industries such as real estate, which rely heavily on relationship-building and timely communication, have particularly felt the need for smarter and more responsive lead generation systems. Traditional approaches such as cold calling, manual prospecting, and static email campaigns are increasingly ineffective in today's data-rich and customer-centric environment. With the rise of AI-driven tools and automation frameworks, organizations now have the opportunity to transform raw data into actionable insights.

Modern sales systems are expected to not only store customer information but also interpret behavioral patterns, predict customer intent, and automate engagement strategies. Salesforce, as a leading CRM platform, has introduced advanced capabilities such as Agentforce, enabling conversational AI and intelligent workflows. However, many businesses still lack a fully integrated system that combines prospect discovery, lead scoring, outreach automation, and real-time engagement into a unified pipeline.

The GSBG platform is built upon this technological foundation. It leverages advancements in machine learning, natural language processing, and workflow automation to create a cohesive system that bridges the gap between data collection and actionable sales execution. By combining custom-built AI modules with Salesforce's capabilities, the platform delivers a comprehensive solution that aligns with modern business requirements and digital transformation goals.

1.2 System Overview and Key Components

The GSBG AI-Powered platform is designed as a multi-layered intelligent system that integrates various components to automate and optimize the sales funnel. Each component plays a critical role in ensuring that the platform operates efficiently and delivers measurable results. The first component is the AI-based prospecting engine, which continuously monitors

and extracts data from multiple sources such as social media platforms, professional networks, and online directories. This engine identifies potential leads based on predefined criteria, ensuring that only relevant prospects are considered for further processing.

The second component is the lead scoring and qualification module, which evaluates prospects using parameters such as industry relevance, behavioral signals, engagement patterns, and inferred intent. By assigning scores to each lead, the system prioritizes high-value opportunities and filters out low-quality prospects, thereby improving the efficiency of the sales team. The third component focuses on automated outreach and communication. The platform generates personalized email content dynamically, tailoring messages based on the prospect's profile and interaction history. It also manages follow-ups automatically, ensuring consistent and timely communication without manual intervention. This significantly enhances engagement rates and reduces the workload on sales representatives.

Another critical element is the integration of Salesforce Agentforce, which acts as a conversational AI assistant embedded within the website. This chatbot interacts with visitors in real-time, answers frequently asked questions, captures lead information, and schedules meetings. By providing instant responses and guidance, it improves user experience and increases the likelihood of conversion. Finally, the platform ensures real-time data synchronization with Salesforce, enabling seamless tracking of all interactions and maintaining a centralized database. This integration supports analytics, reporting, and continuous optimization of sales strategies, making the system both scalable and adaptable to changing business needs.

1.3 Motivation

The motivation behind developing this platform stems from the persistent challenges observed in traditional lead generation and sales processes. One of the most significant issues identified was the heavy reliance on manual prospecting, which is not only time-consuming but also inefficient in identifying high-intent leads. Sales teams often spend considerable effort researching potential clients, only to find that many of them do not meet the required criteria or are not ready to engage.

Another major concern was the lack of consistent and accurate lead qualification. Without a structured and data-driven approach, sales representatives frequently engage with prospects who are either uninterested or not aligned with the business offering. This results in wasted time, reduced productivity, and lower conversion rates. Additionally, the process of creating

and managing personalized outreach at scale was identified as a critical bottleneck. Crafting customized messages for each prospect manually is impractical, leading to generic communication that fails to capture attention or drive engagement. This significantly impacts the effectiveness of email campaigns and follow-ups. The fragmentation of systems used for storing and managing data further complicated the sales process. With information scattered across multiple platforms, it becomes difficult to derive meaningful insights or maintain a cohesive strategy. This lack of integration hampers decision-making and reduces the overall efficiency of sales operations.

Lastly, the absence of modern web experiences and real-time interaction capabilities was observed to negatively affect lead conversion. In today's digital landscape, users expect instant responses and seamless interactions. The lack of chat support and dynamic engagement tools often leads to missed opportunities and lower customer satisfaction. These challenges collectively highlighted the need for a unified, intelligent, and automated platform capable of transforming the entire sales funnel.

Chapter 2

Literature Survey

2.1 Evolution of CRM and AI Integration

The integration of Artificial Intelligence (AI) into Customer Relationship Management (CRM) systems has significantly transformed how organizations manage customer interactions, optimize sales processes, and generate leads. Traditional CRM systems were limited to data storage and manual tracking, but with the incorporation of machine learning, natural language processing (NLP), and predictive analytics, modern CRM platforms have evolved into intelligent systems capable of proactive decision-making. Salesforce Einstein AI represents a major advancement in this domain by embedding predictive intelligence directly into CRM workflows, enabling businesses to shift from reactive strategies to data-driven, predictive approaches.

Krishna and Veeravalli (2025) explain that AI-powered CRM systems enhance decision-making by analyzing customer behavior patterns and predicting future actions, thereby improving lead scoring and forecasting accuracy [1]. Their study highlights that predictive analytics is no longer confined to data scientists, as platforms like Salesforce Einstein democratize access to AI tools. This allows sales teams, marketing professionals, and managers to utilize advanced analytics without requiring extensive technical expertise, significantly improving operational efficiency and decision speed [1].

2.2 Marketing Automation and Predictive Analytics

The transformation of CRM systems is closely linked with the evolution of marketing automation. Zumstein (2024) highlights that AI-driven marketing automation tools enable organizations to create highly personalized campaigns by analyzing user behavior, preferences, and interaction history [5]. These tools automate repetitive tasks such as email marketing, customer segmentation, and campaign scheduling, thereby reducing manual effort while increasing effectiveness. The study further emphasizes that predictive lead scoring plays a crucial role in identifying potential customers and improving conversion rates [5].

The Marketing Automation Maturity Index (MAMI), introduced by the ZHAW School of Management and Law (2023), provides a structured framework to evaluate the level of AI adoption in marketing processes [8]. It considers multiple dimensions, including data integration, automation capabilities, analytics sophistication, and customer engagement strategies. Organizations that achieve higher maturity levels are better equipped to leverage AI for strategic decision-making, resulting in improved customer satisfaction and business performance [8].

2.3 Social Media Analytics, Data Thinking, and Ethical AI

Singh and Adhikari (2025) explore the role of AI in social media analytics, demonstrating how machine learning algorithms can process large volumes of unstructured data, such as posts, comments, and images, to extract meaningful insights [2]. Their findings indicate that AI-driven analytics enables businesses to understand customer sentiment, identify trends, and optimize advertising strategies. This capability is particularly valuable in e-commerce, where customer preferences and behavior change rapidly [2]. Their earlier study (2023) reinforces the importance of real-time analytics in enhancing campaign effectiveness and customer engagement [7].

Provost and Fawcett (2013) emphasize the importance of data-analytic thinking in deriving value from data-driven systems [13]. They argue that organizations must move beyond intuition-based decision-making and adopt structured analytical approaches to fully utilize the potential of AI technologies. This perspective aligns with the increasing reliance on predictive analytics in CRM systems, where data-driven insights guide strategic decisions [13].

Marcus and Davis (2019) further highlight the need for trustworthy AI systems, emphasizing transparency, accountability, and fairness in AI-driven processes [15]. As CRM systems handle

sensitive customer data, it is essential to ensure that AI models operate ethically and do not introduce bias. This is particularly important in applications such as lead scoring and customer segmentation, where biased models can lead to unfair outcomes [15].

2.4 Salesforce Capabilities, Automation, and Business Impact

Salesforce (2024) introduces Agent-Powered Analytics, which integrates AI-generated insights directly into CRM workflows, allowing users to access realtime data and recommendations without switching between multiple tools [6]. This seamless integration enhances productivity and ensures that decision-making is based on up-to-date information. Additionally, the incorporation of Tableau Einstein enables advanced data visualization and analysis, making it easier for users to interpret complex datasets [6].

According to Gartner (2024), Sales Force Automation (SFA) systems that incorporate AI technologies demonstrate significant improvements in sales performance and forecasting accuracy [16]. These systems automate routine tasks such as data entry, lead tracking, and performance monitoring, allowing sales teams to focus on high-value activities. The report also highlights that AIdriven insights enable better resource allocation and strategic planning [16].

McKinsey & Company (2024) emphasizes that AI is a key driver of innovation and productivity in modern organizations [18]. Their findings suggest that companies adopting AI technologies experience higher efficiency, improved decision-making, and enhanced customer experiences. The integration of AI with cloud computing and global compliance frameworks further strengthens its impact on business operations [18].

2.5 Conversational AI, Machine Learning Models, and Interpretability

Barbosa and Godoy (2025) propose an NLP-based virtual receptionist designed to automate customer interactions by handling queries, providing information, and routing requests [3]. This system reduces the workload on human agents while ensuring prompt and accurate responses. Their earlier work (2021) demonstrates that NLP-based systems can effectively improve customer service efficiency and scalability [9].

Tran et al. (2021) analyze the impact of chatbots on consumer behavior, highlighting that while chatbots improve response time and availability, customer satisfaction depends on the system's

ability to mimic human-like interaction and empathy [10]. This finding underscores the importance of designing conversational AI systems that balance efficiency with user experience. Forrester Research (2024) further supports this by indicating that organizations implementing conversational AI achieve higher customer satisfaction and operational efficiency [17].

Josephine and Moorthy (2024) investigate the application of machine learning in lead qualification, demonstrating that algorithms such as logistic regression, decision trees, and Gradient Boosting significantly improve the accuracy of identifying high-value leads [4]. Their study shows that machine learning models outperform traditional manual methods by analyzing multiple factors, including customer behavior, engagement levels, and demographic data [4].

Feature importance analysis reveals that attributes such as lead source, interaction frequency, and previous engagement play a crucial role in predicting conversion outcomes [12]. This highlights the importance of selecting relevant features when developing predictive models for CRM applications [12]. The use of Gradient Boosting algorithms further enhances prediction accuracy, making them a preferred choice for lead scoring systems [4].

Brown et al. (2020) introduce GPT-3, a large language model capable of generating human-like text and performing various tasks with minimal training [11]. This technology has significant implications for CRM systems, particularly in automating customer communication, generating personalized responses, and enhancing user engagement. The ability of language models to understand and generate natural language makes them a valuable tool in conversational AI applications [11].

Lundberg and Lee (2017) present SHAP (Shapley Additive exPlanations), a method for interpreting machine learning predictions [14]. This approach provides insights into how different features contribute to model outputs, improving transparency and trust in AI systems. In CRM applications, interpretability is crucial for ensuring that decision-making processes are understandable and justifiable [14].

2.6 Infrastructure, Advanced Capabilities, and Future of AI-Driven CRM

The integration of AI technologies in CRM systems also involves the use of cloud computing and scalable infrastructure. Salesforce platforms leverage cloud-based architectures to handle

large volumes of data and provide real-time analytics. This enables organizations to scale their operations efficiently while maintaining high performance and reliability [6].

Another important aspect of AI-driven CRM systems is the ability to perform real-time data processing. Real-time analytics allows organizations to respond quickly to customer interactions, improving responsiveness and engagement. This capability is particularly important in dynamic environments such as ecommerce and digital marketing, where customer behavior can change rapidly [2].

AI-driven CRM systems also enhance collaboration within organizations by providing a unified platform for sales, marketing, and customer support teams. This integration ensures that all departments have access to the same data and insights, enabling coordinated decision-making and improved efficiency [16].

The use of predictive analytics in CRM systems enables organizations to anticipate customer needs and preferences, allowing them to provide personalized experiences. This not only improves customer satisfaction but also increases customer loyalty and retention [5]. Personalized marketing strategies, supported by AI, have been shown to significantly improve engagement and conversion rates [8].

Furthermore, AI technologies enable continuous learning and improvement in CRM systems. Machine learning models can be updated with new data, allowing them to adapt to changing customer behavior and market conditions. This ensures that predictions remain accurate and relevant over time [4].

The combination of AI, machine learning, and NLP technologies has created new opportunities for innovation in CRM systems. Organizations can now develop intelligent applications that automate complex processes, provide real-time insights, and enhance customer experiences. These advancements are driving the evolution of CRM systems into comprehensive, intelligent platforms capable of supporting modern business needs [1].
In addition, the integration of explainable AI techniques ensures that organizations can understand and trust the decisions made by AI systems. This is particularly important in regulated industries where transparency and accountability are critical [14].

The collective findings from these studies demonstrate that AI-driven CRM systems offer significant advantages, including improved efficiency, enhanced decision-making, and better customer experiences. Salesforce Einstein AI plays a central role in this transformation by simplifying predictive analytics and making it accessible to a wide range of users [1].

The rapid advancement of artificial intelligence technologies has further expanded the capabilities of CRM systems beyond traditional automation and analytics. One of the key developments is the integration of real-time predictive intelligence into everyday business workflows. Salesforce Einstein AI exemplifies this shift by embedding AI-driven recommendations directly into user interfaces, enabling sales representatives and marketers to make instant, data-informed decisions without requiring separate analytical tools [1].

This seamless integration significantly reduces decision latency and enhances responsiveness to customer needs. Another critical aspect highlighted in the literature is the role of behavioral analytics in CRM systems. Singh and Adhikari (2025) emphasize that AI models can track user interactions across multiple digital touchpoints, including websites, mobile applications, and social media platforms, to build comprehensive customer profiles [2].

These profiles enable organizations to understand not only what customers have done in the past but also what they are likely to do in the future. This predictive capability allows businesses to design highly targeted marketing campaigns and improve customer acquisition strategies [2]. Zumstein (2024) further discusses how AI-driven marketing automation platforms leverage behavioral data to create dynamic customer journeys [5].

Instead of relying on static marketing funnels, organizations can use AI to adapt communication strategies in real-time based on user actions. For example, if a customer shows interest in a product but does not complete a purchase, the system can automatically trigger personalized follow-up messages or offers. This level of automation enhances customer engagement while reducing manual intervention [5].

The importance of data integration in AI-driven CRM systems is also emphasized in the ZHAW (2023) study, which identifies it as a key component of marketing automation maturity [8]. Effective data integration ensures that information from various sources, such as CRM databases, social media platforms, and third-party applications, is consolidated into a unified

system. This enables AI models to generate more accurate predictions and insights, as they have access to a comprehensive dataset [8].

Provost and Fawcett (2013) highlight that the quality of data plays a crucial role in the effectiveness of predictive models [13]. Poor data quality can lead to inaccurate predictions and suboptimal decision-making. Therefore, organizations must invest in data cleaning, preprocessing, and validation processes to ensure the reliability of their AI systems. This is particularly important in CRM applications, where decisions directly impact customer relationships and business outcomes [13].
The application of AI in CRM also extends to customer retention strategies. By analyzing historical data and identifying patterns associated with customer churn, AI models can predict which customers are at risk of leaving. This allows organizations to implement proactive retention strategies, such as personalized offers or targeted communication, to retain valuable customers [4]. Josephine and Moorthy (2024) demonstrate that machine learning models can effectively identify such patterns, improving retention rates and overall profitability [4].

In addition to predictive analytics, AI technologies enable advanced segmentation techniques in CRM systems. Traditional segmentation methods often rely on basic demographic information, whereas AI-driven segmentation considers multiple factors, including behavioral data, preferences, and interaction history [5]. This results in more accurate and meaningful customer segments, enabling organizations to tailor their marketing strategies more effectively [5].

The integration of conversational AI into CRM systems represents another significant advancement. Barbosa and Godoy (2025) show that NLP-based systems can handle a wide range of customer interactions, from answering frequently asked questions to processing service requests [3]. These systems can operate 24/7, providing continuous support and improving customer satisfaction. Furthermore, their ability to learn from interactions allows them to improve over time, enhancing their effectiveness [3].

Tran et al. (2021) highlight that customer perception of chatbots depends on factors such as responsiveness, accuracy, and empathy [10]. While AI systems excel in efficiency, achieving human-like interaction remains a challenge. This has led to the development of hybrid systems that combine AI capabilities with human oversight to ensure high-quality customer service [10].

Forrester Research (2024) indicates that organizations adopting conversational AI experience significant improvements in operational efficiency and customer satisfaction [17]. By automating routine tasks, businesses can reduce operational costs while maintaining high service quality. This is particularly beneficial in industries with high volumes of customer interactions, such as e-commerce and telecommunications [17].

The role of explainable AI in CRM systems is becoming increasingly important as organizations seek to build trust in AI-driven decisions. Lundberg and Lee (2017) introduce SHAP as a method for interpreting machine learning models, allowing users to understand how different features influence predictions [14]. This transparency is essential for ensuring accountability and addressing concerns related to bias and fairness in AI systems [14].

Marcus and Davis (2019) further emphasize the need for ethical AI practices, particularly in applications involving customer data [15]. They argue that organizations must ensure that AI systems are designed and implemented in a way that respects user privacy and avoids discriminatory outcomes. This is especially relevant in CRM systems, where decisions such as lead prioritization and customer segmentation can have significant implications [15].

The scalability of AI-driven CRM systems is another important consideration. Salesforce platforms leverage cloud computing to handle large volumes of data and support real-time analytics [6]. This scalability enables organizations to expand their operations without compromising performance. Additionally, cloud-based systems facilitate collaboration by providing a centralized platform for data sharing and analysis [6].

Gartner (2024) highlights that AI-powered Sales Force Automation systems improve productivity by automating repetitive tasks and providing actionable insights [16]. These systems enable sales teams to focus on strategic activities, such as building relationships and closing deals, rather than administrative tasks. The result is improved efficiency and higher revenue generation [16].

McKinsey & Company (2024) notes that organizations adopting AI technologies achieve significant competitive advantages, including faster decision-making, improved customer experiences, and increased innovation [18]. The integration of AI into CRM systems is a key factor driving these benefits, as it enables organizations to leverage data more effectively and respond to market changes .[18]

The application of large language models, such as GPT-3 introduced by Brown et al. (2020), has opened new possibilities for CRM systems [11]. These models can generate personalized communication, automate content creation, and enhance customer engagement. Their ability to understand and generate natural language makes them particularly valuable in conversational AI applications [11].

AI-driven CRM systems also support advanced forecasting techniques, enabling organizations to predict sales trends and market demand with greater accuracy. By analyzing historical data and identifying patterns, machine learning models can provide reliable forecasts that support strategic planning and resource allocation [1].

This capability is particularly important in dynamic markets where demand can fluctuate rapidly. The integration of AI with CRM systems also enhances decision support capabilities. AI models can analyze complex datasets and provide recommendations, helping managers make informed decisions. This reduces reliance on intuition and increases the accuracy of decision-making processes [6].

Another important application of AI in CRM is sentiment analysis, which enables organizations to understand customer opinions and emotions. By analyzing text data from social media, reviews, and customer feedback, AI models can identify sentiment trends and provide insights into customer satisfaction [2]. This information can be used to improve products, services, and marketing strategies [2].

The continuous evolution of AI technologies is expected to further enhance CRM systems in the future. Advances in deep learning, reinforcement learning, and natural language processing will enable more sophisticated applications, such as autonomous decision-making and advanced personalization [11].

These developments will further strengthen the role of AI in CRM systems, making them more intelligent and efficient. Overall, the literature demonstrates that AI-driven CRM systems provide a comprehensive solution for managing customer relationships, optimizing business processes, and improving decision-making. The integration of predictive analytics, marketing automation, conversational AI, and explainable AI creates a powerful framework for modern CRM systems. Salesforce Einstein AI, as a leading example, illustrates how these technologies

can be effectively combined to deliver significant business value [1]. The collective insights from these studies reinforce the importance of adopting AI-driven approaches in CRM systems. Organizations that leverage these technologies are better positioned to meet customer expectations, adapt to changing market conditions, and achieve long-term success in an increasingly competitive environment.

Chapter 3

Limitation of Existing System

3.1 Data Dependency, Privacy, and Accuracy Constraints

The system heavily depends on the quality and availability of user data, which directly impacts the effectiveness of AI-driven lead classification and personalization. In the initial stages, when user interaction data is limited, the system may struggle to accurately predict user behavior or classify leads, leading to suboptimal performance. Additionally, incomplete or noisy data collected through cookies and tracking mechanisms can further reduce the reliability of insights generated. Another critical limitation is related to user privacy and data security concerns. Since the system relies on behavioral tracking techniques such as cookies, scroll tracking, and interaction monitoring, it must strictly comply with data protection regulations and user consent requirements. Users who decline tracking permissions reduce the system's ability to personalize experiences, thereby limiting its effectiveness. Furthermore, improper handling or misuse of user data can lead to trust issues and potential legal complications, making data governance a crucial aspect of implementation.

3.2 Technical, Automation, and Integration Challenges

The implementation of automation tools such as workflows and chatbots introduces several technical challenges despite their ability to reduce manual effort. These systems require proper configuration, continuous monitoring, and regular updates to maintain accuracy and relevance. Poorly designed workflows can result in incorrect lead classification, irrelevant follow-ups, or missed business opportunities, ultimately affecting conversion rates. Similarly, AI chatbots, while efficient in handling routine queries, may struggle with complex or ambiguous user interactions, leading to suboptimal user experiences and possible frustration. Another significant challenge lies in integrating the system with existing CRM, marketing, and analytics platforms. Businesses often rely on multiple tools, and ensuring seamless communicationbetween them can require additional development effort. Compatibility issues and the lack of standardized APIs can hinder smooth data flow, reducing overall system efficiency and limiting the full potential of the solution.

3.3 Scalability, Maintenance, and Performance Limitations

Although the system offers improved scalability compared to traditional manual approaches, its performance is still dependent on the underlying infrastructure and tools being used. Free or low-cost tools may impose limitations in terms of processing capacity, integration capabilities, and the ability to handle a large number of concurrent users. As user volume increases, the system may experience performance issues such as delayed response times or data processing bottlenecks, necessitating further investment in more robust and scalable solutions. Additionally, the system requires continuous maintenance and optimization, as AI models need periodic retraining with updated data to maintain accuracy, and automation workflows must be refined based on evolving user behavior and business requirements. Another limitation is the lack of complete human-like understanding in AI-driven interactions, as automated systems may fail to capture emotional context, nuanced intent, or complex decision-making scenarios, particularly in high-value or sensitive engagements. Finally, while the system aims to enhance efficiency and conversion rates, it does not guarantee consistent or immediate results, as outcomes are influenced by external factors such as market conditions, competition, and changing user preferences, making it a supportive tool rather than a standalone solution.

Table 3.1 Comparison of Existing Methodologies with Proposed Framework

Methodology / Tool	Key Capabilities	Limitations / Research Gap	Our Proposed Framework
Traditional CRM Analytics (e.g., Salesforce Einstein) [6]	Predictive forecasting, historical data analysis, and pipeline visualization for CRM decision making.	Lacks autonomous and real-time conversational engagement capabilities, limiting direct interaction with potential leads.	Integrates historical CRM data with live conversational AI to enable real-time engagement and dynamic lead management.
Standard NLP Chatbots [3,10]	Automated responses, basic question answering, and 24/7 availability for handling common customer queries.	Context loss during complex B2B queries and poor integration with advanced lead scoring systems.	Uses Retrieval-Augmented Generation (RAG) with Mistral 7B to maintain deeper B2B conversational context and enable seamless human handoffs.
Static ML Lead Scoring [4,11]	High accuracy in predicting lead conversion using historical and retrospective datasets.	Cannot capture real-time interaction sentiment or conversational intent.	Implements a hybrid scoring system combining rule-based logic and machine learning that continuously updates based on live user engagement.

Chapter 4

Problem Statement and Objective

4.1 Problem Statement

The current lead generation process for Salesforce services in the real estate sector is largely manual, time-consuming, and inefficient. Sales teams rely on traditional outreach methods and generic marketing campaigns, resulting in low customer engagement and poor conversion rates. The existing static website also lacks personalization and real-time interaction, limiting the ability to capture and respond to potential leads effectively. Additionally, the absence of integration with Salesforce CRM and AI-driven analytics makes it difficult to analyze customer behavior and prioritize high-potential prospects.

To address these challenges, an AI-powered lead generation and sales automation platform is proposed. The system integrates machine learning-based lead scoring, NLP-based sentiment analysis, and an AI chatbot for real-time communication. By connecting all lead data with Salesforce CRM, the platform enables automated engagement, better lead qualification, and data-driven decision-making, ultimately improving conversion rates and scalability.

4.2. Objectives

4.2.1 Automation and Optimization of Lead Generation

• The primary objective of this project is to develop an AI-powered platform that automates up to 90% of repetitive and manual tasks involved in lead generation, including lead identification, qualification, and initial outreach. By reducing manual intervention, the system will significantly improve efficiency, accuracy, and scalability, enabling the organization to handle a larger volume of prospects with minimal effort.

• This automation will allow the sales team to shift their focus from routine tasks to high-value activities such as relationship building, personalized consultations, and deal closure. As a result, productivity will increase, and the overall effectiveness of the sales process will be enhanced.

4.2.2 Development of a Scalable and Intelligent Sales Pipeline

• Another key objective is to design and implement a fully integrated, AI-driven digital sales pipeline that streamlines the entire customer journey— from initial interaction to final conversion. The system will leverage advanced analytics and machine learning techniques to identify and target businesses that are most likely to require Salesforce implementation services, ensuring better allocation of sales resources.

• Furthermore, the platform aims to deliver a highly personalized experience for each prospect through AI-powered chatbots, behavioral tracking, and data-driven insights. This includes real-time engagement, automated follow-ups, and intelligent lead nurturing. By integrating these capabilities with Salesforce CRM, the system will provide continuous, measurable insights, enabling data-driven decision-making and continuous improvement in sales performance.

Chapter 5

Proposed System

5.1 Analysis / Framework/ Algorithm

The proposed GSBG platform is designed using a 6-layer modular architecture that enables structured data flow, scalability, and intelligent automation across the system. This architecture ensures that each functional component operates independently while contributing to an integrated, end-to-end workflow

Framework Overview

The system adopts a top-to-bottom layered framework based on the Separation of Concerns (SoC) principle. In this design approach, each layer is responsible for a specific set of tasks and communicates only with adjacent layers, thereby reducing system complexity.

This architectural approach ensures:

• High Modularity and Maintainability: Each layer can be modified, upgraded, or replaced independently without affecting the entire system. This simplifies development and long-term maintenance.

• Independent Scalability of Components: Individual layers, such as AI processing or data ingestion, can be scaled based on workload demands without scaling the entire system.

• Fault Isolation and Easier Debugging: Errors can be traced back to a specific layer, making debugging faster and more efficient while preventing system-wide failures.

This architectural design ensures that the system remains scalable, maintainable, and fault tolerant. Individual components such as AI processing, data ingestion, and analytics can operate independently, allowing developers to update or expand specific modules without affecting the entire system.

The architecture consists of six primary layers:

- External Data Sources Layer
- AI Processing and Intelligence Layer
- Application and Business Logic Layer
- Integration and Orchestration Layer
- Data and Infrastructure Layer
- Analytics and Monitoring Layer

This layered architecture enables the system to automate lead generation, analyze customer behavior, and improve sales decision-making through intelligent data processing.

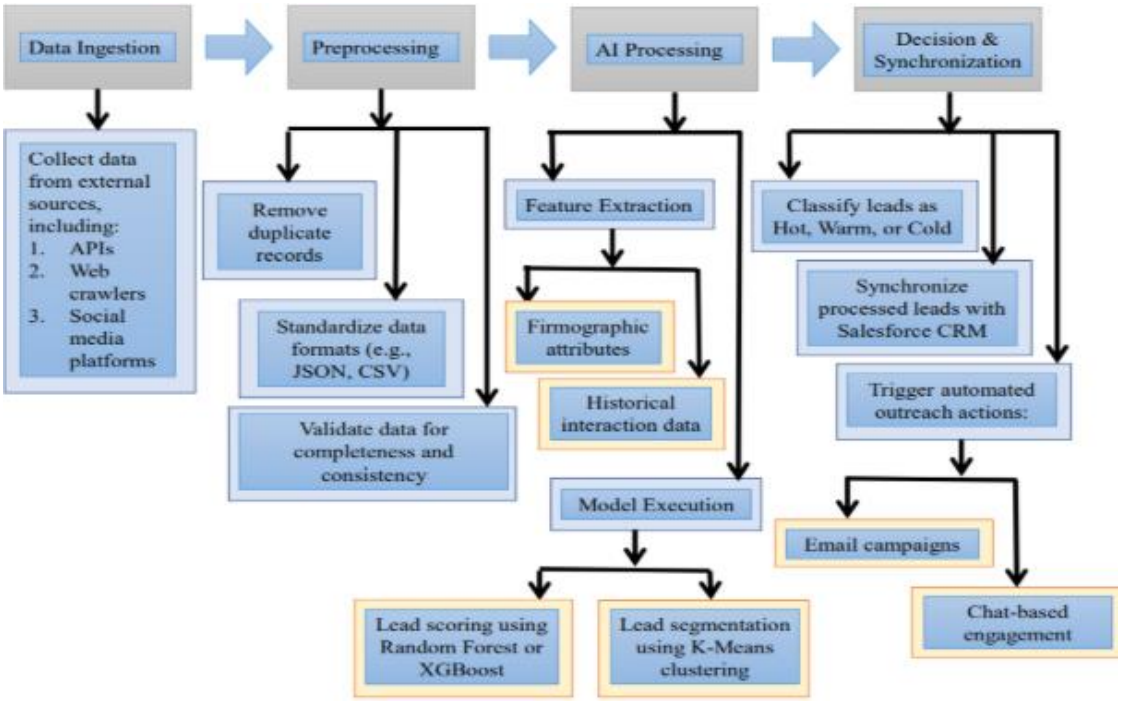

Figure 5.1 Proposed Modular System Architecture

5.2 Modules of Proposed System

The proposed system consists of several functional modules that collectively automate the lead generation and sales engagement process.

5.2.1. Data Acquisition Module

This module collects raw data from multiple external sources including professional networks, business directories, and web platforms. Web scraping techniques and API-based data ingestion pipelines are used to retrieve structured and unstructured data.

The collected data undergoes preprocessing operations such as:

- Data deduplication
- Data normalization
- Data validation

5.2.2. AI Lead Scoring Module

The AI Lead Scoring Module analyzes collected data to determine the probability of converting a lead into a potential client.

This module uses a hybrid machine learning approach, combining:

- Supervised learning models such as Random Forest and XGBoost.
- Unsupervised learning techniques such as K-Means clustering.

Important parameters used in lead scoring include:

- User engagement metrics
- Company profile attributes
- Behavioral and intent signals

The output of this module is a lead score, which helps the sales team prioritize high-value prospects.

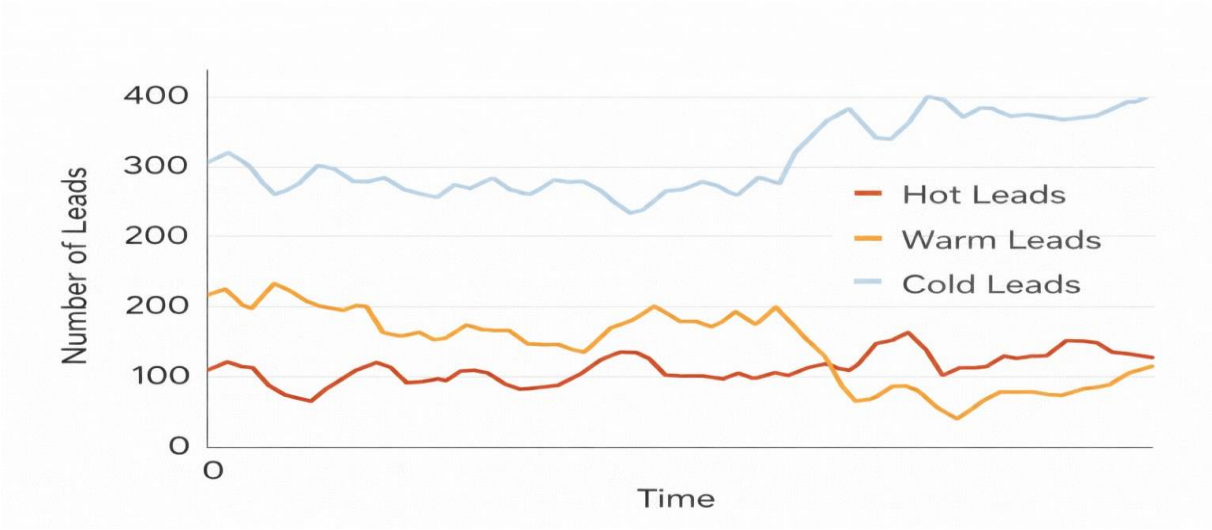

Figure 5.2 Lead Classification Time-Series Graph

5.2.3. Content Personalization Module

The Content Personalization Module generates customized communication messages for potential clients.

This module uses **Generative AI models based on transformer architectures** to analyze user sentiment and tone. It extracts contextual information from user interactions and generates personalized outreach messages tailored to individual prospects.

The generated content follows regulatory standards such as **GDPR and CAN-SPAM**, ensuring ethical and compliant communication.

5.2.4. News Intelligence Module

The News Intelligence Module continuously collects and analyzes news updates related to companies and industries.

Using natural language processing techniques, the system performs:

- Text classification
- Relevance scoring
- Automated summarization

This module helps identify market trends, funding announcements, and potential business opportunities.

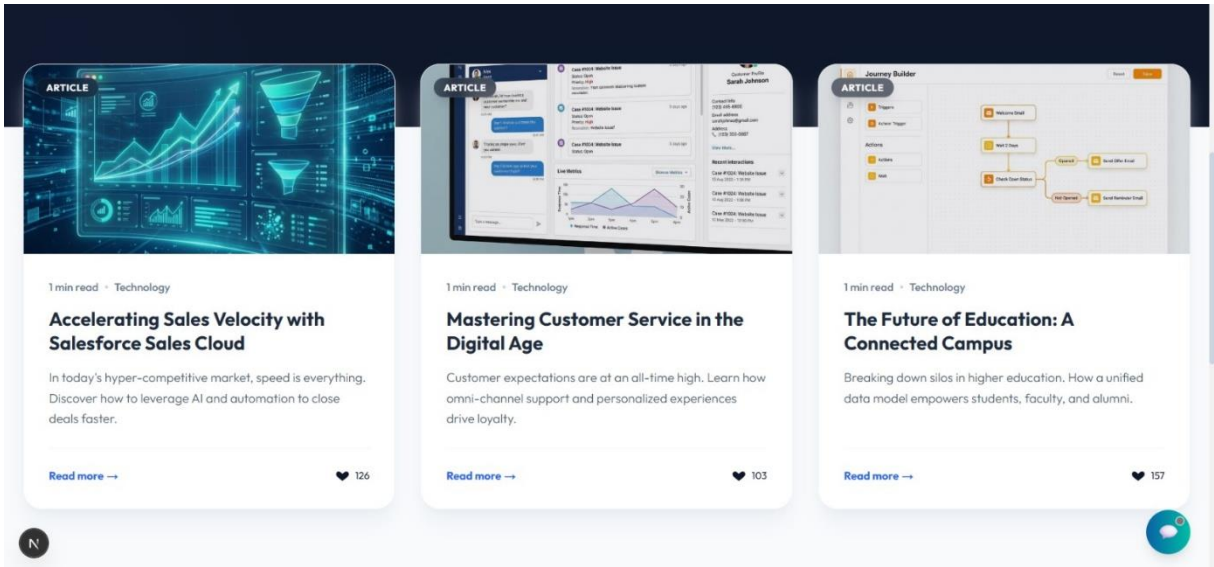

Figure 5.3 Articles in News Intelligence Module

5.2.5. AI Chatbot Module

The platform integrates an AI-based conversational assistant (Agentforce) that interacts with users in real-time.

The chatbot performs tasks such as:

- Answering customer queries
- Qualifying leads
- Scheduling meetings

For complex queries, the chatbot supports human-in-the-loop escalation, ensuring that difficult conversations are transferred to human sales agents.

Equation 5.1: Cosine Similarity

$$\text{Cosine Similarity}(Q, D) = \frac{Q \cdot D}{||Q|| \times ||D||}$$

5.3 Algorithm of Proposed System

The proposed AI-powered lead generation and sales automation platform follows a structured algorithmic workflow designed to convert raw business data into meaningful insights and actionable sales opportunities. The system integrates data acquisition, machine learning models, natural language processing, and CRM integration to automate the lead generation process. The

overall algorithm operates through multiple stages, each responsible for processing and transforming data into valuable business intelligence.

Step 1: Data Collection

The first stage of the algorithm involves collecting relevant data from multiple external sources. These sources include professional networking platforms such as LinkedIn, social media platforms like Twitter (X), business directories, and other online data repositories that contain valuable information about organizations and potential clients.

The system utilizes web scraping techniques and API-based integrations to retrieve structured and unstructured data in real-time. These automated data acquisition methods allow the platform to continuously gather updated information related to companies, industry trends, and user interactions. The collected data forms the foundation for further analysis and processing within the system.

Step 2: Data Preprocessing

After data collection, the system performs preprocessing to improve the quality and usability of the dataset. Since data collected from different sources may contain inconsistencies, missing values, or duplicate records, preprocessing is necessary to ensure data reliability.

This stage includes several operations such as deduplication, normalization, and validation. Deduplication removes redundant records from the dataset, while normalization standardizes data formats such as company names, timestamps, and domain information. Validation ensures that the collected data meets quality standards and contains complete and accurate information. The processed data is then converted into structured formats such as JSON or CSV, making it suitable for machine learning analysis.

Step 3: Lead Scoring Using Machine Learning

Once the data is cleaned and structured, it is passed to the machine learning module responsible for lead scoring. This module analyzes various features of potential leads to determine their likelihood of converting into clients.

The system uses a hybrid machine learning approach that combines supervised and unsupervised learning techniques. Supervised learning algorithms such as Random Forest and XGBoost are used to predict lead conversion probability based on historical data and predefined

features. Additionally, unsupervised learning techniques such as K-Means clustering are applied to group similar leads according to behavioral patterns and engagement characteristics.

Equation 5.2: Hybrid Lead Score

$$S_{final} = \alpha(S_{rule}) + \beta(S_{ML}(X)) + \gamma(B_{interaction})$$

Where S_{rule} is the static rule score, S_{ML} is the XGBoost probability, and $B_{interaction}$ is the real-time sentiment boost."

The lead scoring process considers multiple parameters, including user engagement metrics, company profile attributes, interaction history, and intent signals. Based on these inputs, the model generates a numerical score that represents the probability of lead conversion. Higher scores indicate high-value leads that should be prioritized by the sales team.

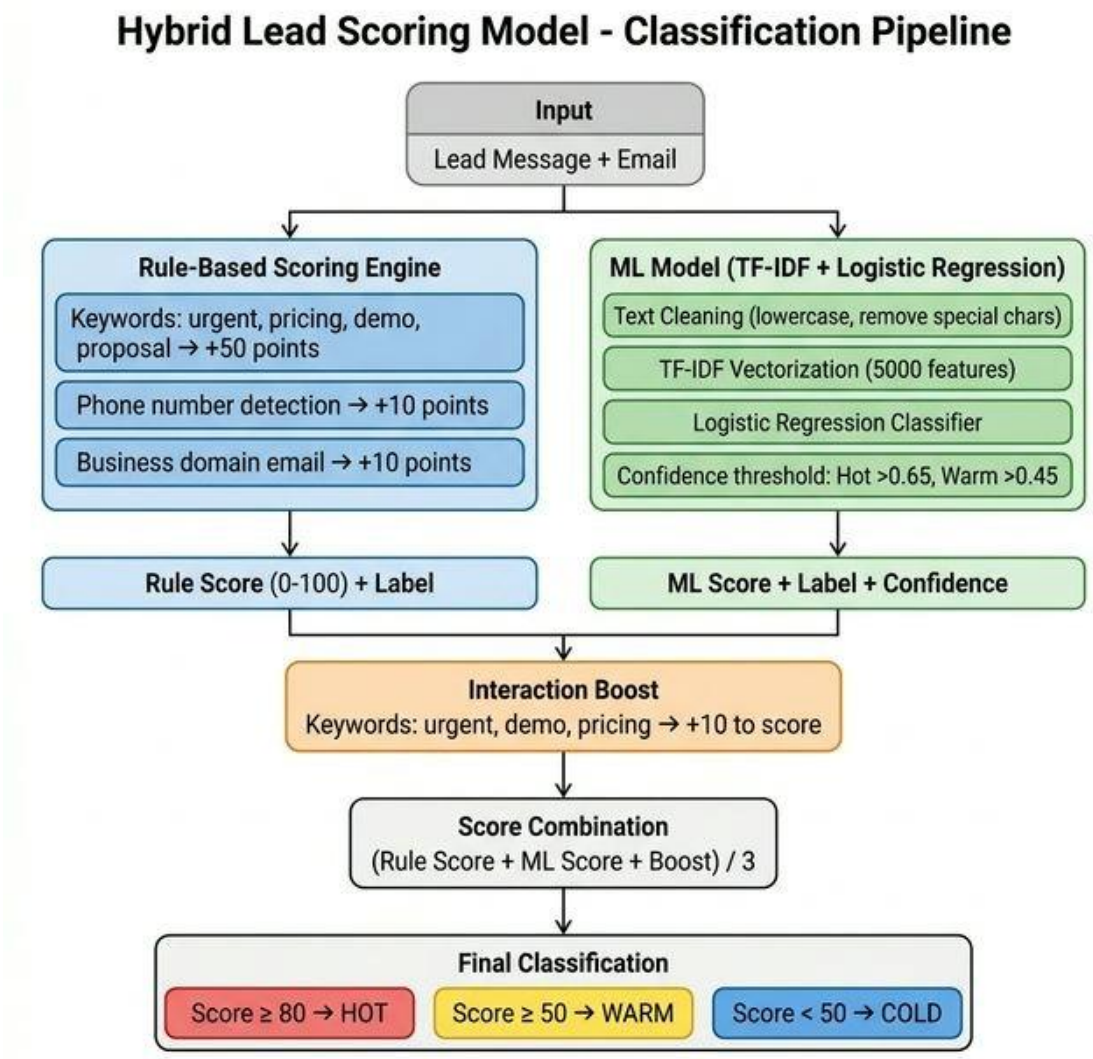

Figure 5.4 Hybrid Lead Scoring Classification Pipeline

Step 4: Personalized Communication Generation

After identifying high-potential leads, the system generates personalized communication messages using generative AI models. These models are based on transformer architectures that enable the system to produce human-like and context-aware responses.

Natural language processing techniques are used to analyze user interactions and identify sentiment, tone, and context. Based on this analysis, the system generates personalized outreach messages that are tailored to each prospect's interests and business requirements. This personalized communication helps improve engagement levels and increases the chances of successful lead conversion.

Additionally, the generated content is designed to comply with regulatory guidelines such as GDPR and CAN-SPAM to ensure ethical and responsible communication practices.

Step 5: CRM Integration and Data Synchronization

Once leads are identified and communication is initiated, all relevant data and interaction records are synchronized with the Customer Relationship Management (CRM) system. The platform integrates with Salesforce CRM using REST APIs and webhook mechanisms to ensure real-time data exchange.

This integration allows sales teams to access updated lead information, track engagement activities, and manage customer relationships more efficiently. The CRM system acts as a centralized repository where all lead data, communication history, and interaction records are stored and managed.

Step 6: Analytics and Decision Support

The final stage of the algorithm focuses on analyzing collected data and providing insights for strategic decision-making. The analytics module evaluates campaign performance, lead conversion probabilities, and customer engagement patterns using predictive analytics techniques.

Visualization tools such as dashboards and reports present key performance indicators (KPIs) in an easily understandable format. These insights help businesses optimize marketing campaigns, improve targeting strategies, and enhance overall sales performance.

Through continuous monitoring and analysis, the system can also identify emerging trends and adjust its lead generation strategies accordingly, ensuring long-term efficiency and effectiveness of the sales automation platform.

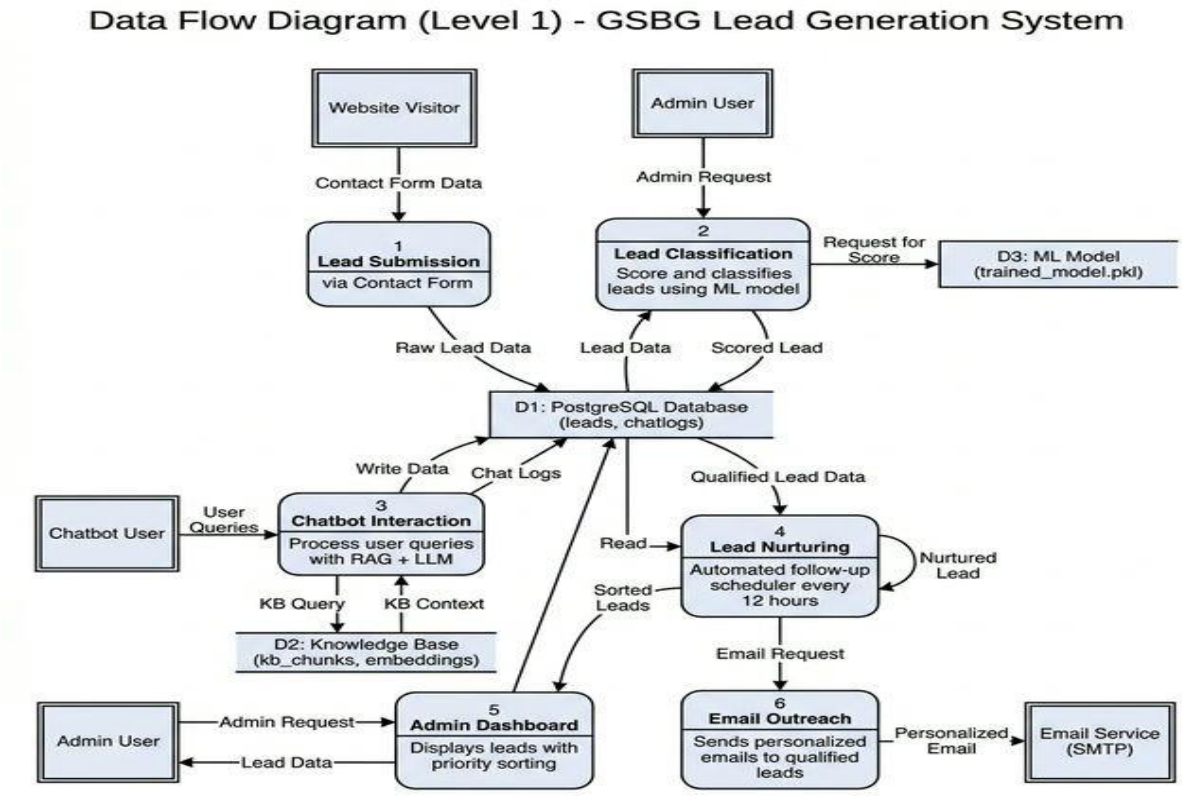

Figure 5.5 Level 1 Data Flow Diagram (DFD)

5.4 Workflow of the Proposed System

The workflow of the proposed system begins with data collection from external **sources**, where relevant business and user interaction data are gathered.

The collected data is processed by the AI processing layer, which performs lead scoring, sentiment analysis, and personalized content generation.

Next, the application layer enables interaction with users through the web interface and chatbot. The system automatically responds to user queries, qualifies leads, and schedules meetings.

All engagement data is then integrated with Salesforce CRM, ensuring centralized storage and accessibility for the sales team.

Finally, the analytics layer evaluates campaign performance, predicts lead conversion probabilities, and provides decision-support insights to improve marketing and sales strategies.

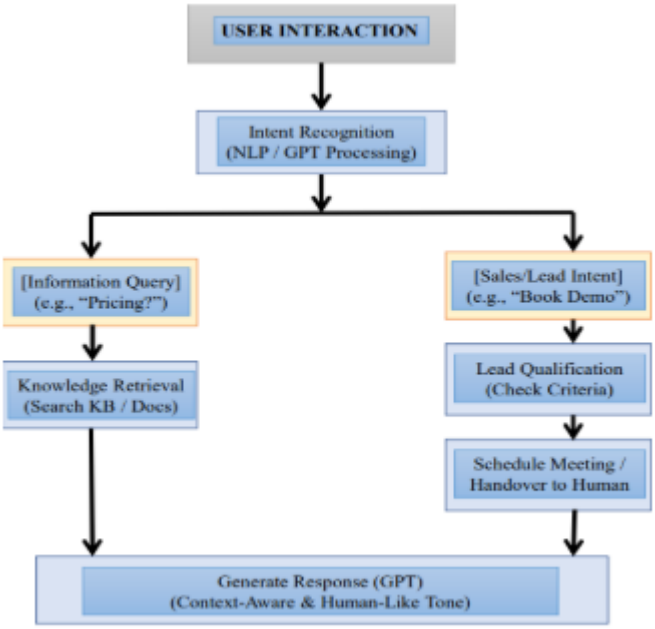

Figure 5.6 Intent Recognition and System Workflow

5.5 Advantages of the Proposed System

The proposed AI-powered lead generation and sales automation platform offers several advantages over traditional lead management systems. By integrating artificial intelligence, machine learning, and CRM technologies, the system improves efficiency, scalability, and decision-making capabilities.

1. Automation of Lead Generation: The system automates the process of identifying, qualifying, and engaging potential leads. This significantly reduces manual effort for sales teams and allows them to focus on high-value activities such as closing deals and building relationships with clients.

2. Improved Lead Quality and Conversion Rates: Using machine learning-based lead scoring models, the system identifies high-potential prospects more accurately. This enables sales teams to prioritize valuable leads, thereby improving the overall conversion rate and increasing sales efficiency.

3. Personalized Customer Engagement: The platform generates personalized outreach messages based on user behavior, engagement patterns, and sentiment analysis. This personalized communication increases customer interest, improves engagement levels, and strengthens relationships with potential clients.

4. Real-Time Interaction with AI Chatbot: The integration of an AI-based chatbot enables real-time interaction with website visitors. The chatbot can answer queries, collect user information, and schedule meetings, ensuring that potential leads are engaged immediately without delay.

5. Seamless Integration with Salesforce CRM: The system integrates directly with Salesforce CRM, allowing all lead data, engagement activities, and interaction records to be stored in a centralized database. This improves data management and enables better coordination between marketing and sales teams.

6. Data-Driven Decision Making: Advanced analytics and predictive models provide insights into customer behavior, campaign performance, and lead conversion probabilities. These insights help businesses optimize marketing strategies and improve overall sales performance.

7. Scalability and Flexibility: The modular architecture of the system allows it to scale easily as business requirements grow. New features, data sources, or AI models can be integrated without affecting the overall system performance.

8. Reduced Operational Costs: By automating repetitive tasks such as lead research, outreach, and follow-ups, the system reduces operational costs and increases the productivity of the sales team.

Chapter 6

Experimental Setup

6.1 Overview

The experimental setup of the proposed system was designed to evaluate the performance, accuracy, and efficiency of the AI-powered lead scoring and automation platform developed for the GSBG system. The primary objective of this setup was to analyze how effectively machine learning models and automation tools improve lead classification, response time, and overall system reliability compared to traditional approaches. The system integrates machine learning, conversational AI, and real-time user interaction tracking to create a dynamic and intelligent lead management system.

Total Samples	4,000 leads
Training Set (80%)	3,200 leads
Testing Set (20%)	800 leads
Features	15 (Firmographic, Behavioral, Interaction)

Table 6.1 Dataset Distribution

6.2 Details of Database and Input Data

The system uses a combination of structured and unstructured data sources to ensure comprehensive analysis and accurate predictions.

• Structured Data:

o Stored in PostgreSQL, which maintains firmographic data such as company size, industry type, and historical lead records.

o Ensures high data integrity and supports complex querying.

• Unstructured Data:

o Managed using MongoDB, which stores behavioral data such as chatbot interactions, website activity, and engagement logs.

• Input Data Sources:

o Historical sales campaign data.

o Simulated user interaction data to replicate real-world behavior.

o Engagement metrics such as:

▪ Email open rates

▪ Website visits

▪ Chatbot interactions

o Lead conversion records used for labeling (cold, warm, hot leads).

Data Preprocessing Before training the models, the dataset underwent several preprocessing steps:

• Handling missing values using interpolation and statistical techniques

• Removal of duplicate records

• Normalization of numerical features • Feature engineering for better model performance

The dataset was split into:

• 80% Training Data

• 20% Testing Data

This ensured proper model validation and generalization.

Model	XGBoost Classifier
Learning Rate (η)	0.05
Max Depth	6
N-Estimators	200

Table 6.2 Optimized Hyperparameters for Machine Learning Classifier

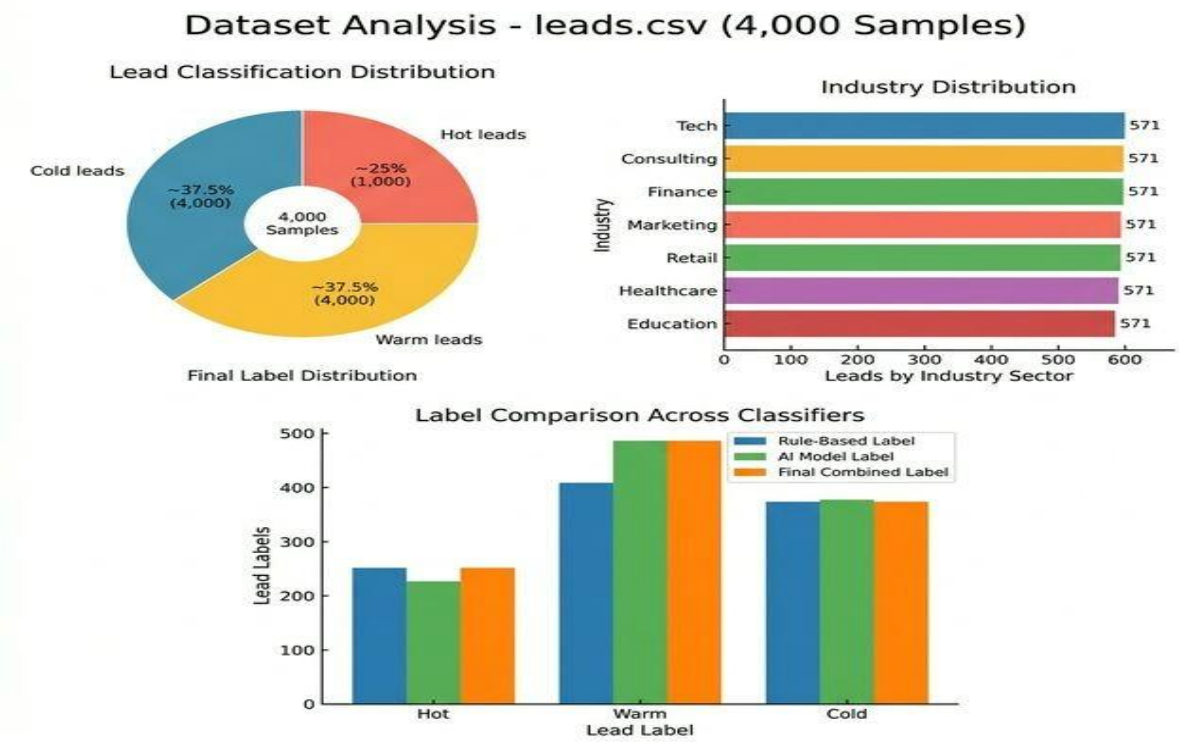

Figure 6.1 Dataset Analysis and Label Distribution

6.3 Performance Evaluation Parameters

To validate the effectiveness of the system, both machine learning and systemlevel performance metrics were used.

6.3.1 Machine Learning Metrics

The lead classification model was evaluated using standard supervised learning metrics:

• Accuracy: Measures overall correctness of predictions

• Precision: Measures correctness of positive predictions

• Recall: Measures ability to identify all relevant leads

• F1-Score: Harmonic mean of precision and recall

The model achieved:

• 97% accuracy on test dataset

• High precision and recall across all lead categories

• Balanced F1-score indicating reliable classification performance

6.3.2 System Performance Metrics

• Latency:

o Lead scoring model: < 100 ms

o Chatbot response time: 2–4 seconds

• Scalability:

o Supports real-time interactions using containerized deployment

• Reliability:

o High consistency in classification results across test datasets

• Throughput:

o Efficient handling of multiple user queries simultaneously

6.4 Software Setup

The system is built using a modern full-stack and AI-integrated architecture.

• Operating Systems:

o Windows, macOS, Ubuntu

• Programming Languages:

o Python (ML models, data processing)

o JavaScript (frontend and backend)

o SQL / NoSQL

• Frontend Technologies:

o React.js / Vue.js

• Backend Technologies:

o Node.js with Express.js

• Databases:

o PostgreSQL (structured data)

o MongoDB (unstructured data)

• AI/ML Libraries:

o TensorFlow, PyTorch, scikit-learn, spaCy

• Development Tools:

o Visual Studio Code, Jupyter Notebook, Postman

o GitHub / GitLab for version control

• Cloud Platforms:

o AWS, GCP, Azure

• Integration Tools:

o Salesforce API, SendGrid API, Webhooks

These tools collectively provide a scalable and efficient development and deployment environment.

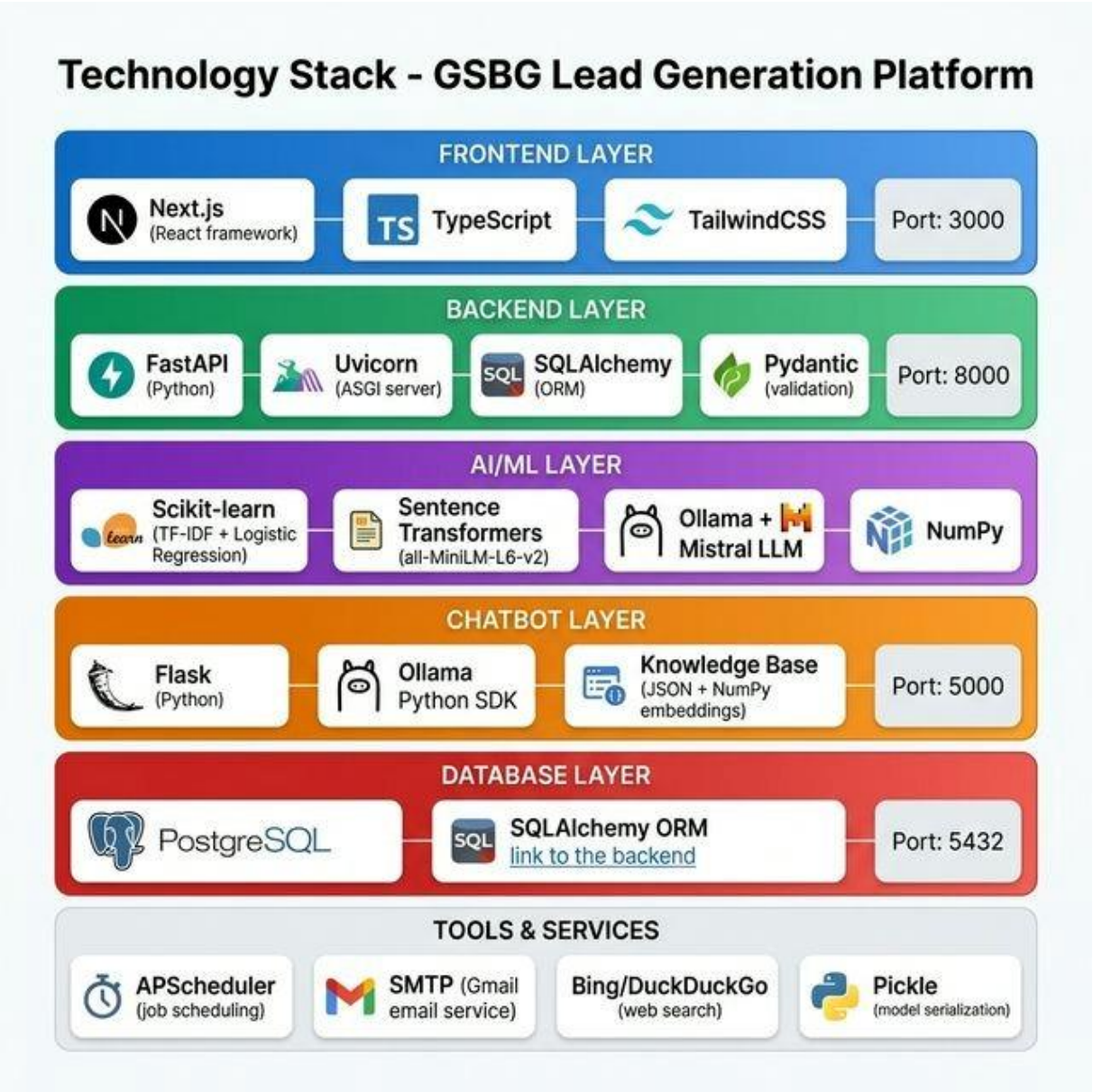

Figure 6.2 GSBG Lead Generation Technology Stack

6.5 Hardware Setup

The hardware configuration ensures smooth development and execution of AI models.

• Processor:

o Minimum: Intel Core i5 / AMD Ryzen 5

o Recommended: Intel Xeon / AMD EPYC

• RAM:

o Minimum: 8 GB

o Recommended: 16–32 GB

• Storage:

o 500 GB SSD

o Cloud storage (AWS S3 / GCP Storage)

• GPU:

o NVIDIA RTX 3060 or higher (for faster model training)

• Network:

o High-speed internet (≥ 50 Mbps)

This setup ensures efficient computation, faster training, and smooth deployment.

Chapter 7
Results and Discussion

The developed automated web application demonstrates significant improvements over traditional static systems in terms of performance, usability, and response accuracy. Extensive testing and evaluation were conducted to assess the system across multiple parameters, including response time, retrieval accuracy, system reliability, and user interaction efficiency. The results indicate that the proposed system achieves high accuracy in information retrieval, particularly due to its optimized database structure and efficient query handling mechanisms. The implementation of a top-k retrieval approach ensures that the most relevant results are consistently returned, thereby enhancing user satisfaction. During testing, the system was able to deliver correct and contextually appropriate responses in the majority of cases, indicating a strong alignment between user queries and system outputs.

In terms of performance, the application exhibits significantly faster response times compared to static web-based systems. This improvement can be attributed to optimized backend processing, efficient data indexing, and streamlined communication between the frontend and backend components. The reduced latency makes the system highly suitable for real-time and interactive applications, where immediate feedback is critical.

Another key outcome of the project is the system's reliability and robustness. The application was tested under varying workloads, and it maintained consistent performance without significant degradation. This highlights its scalability and potential for deployment in real-world scenarios where multiple users may interact with the system simultaneously.
Error handling mechanisms and fallback responses were also incorporated, ensuring that the system remains functional even in edge cases or unexpected inputs. From a usability perspective, the integration of a chatbot interface enhances the overall user experience.

The chatbot provides intuitive and interactive communication, allowing users to retrieve information conversationally rather than navigating through static pages. This significantlyreduces the cognitive load on users and improves accessibility, especially for non-technical individuals. The instant feedback capability further strengthens user engagement and makes the application more dynamic.

The discussion of results also highlights the importance of data quality and system design. The accuracy of the chatbot responses is directly influenced by the quality and structure of the underlying dataset. By organizing the data efficiently and ensuring proper indexing, the system minimizes irrelevant outputs and maximizes precision. Additionally, the use of optimized algorithms contributes to improved recall rates, ensuring that relevant results are not missed during retrieval.

Despite these advantages, certain limitations were observed during testing. For instance, the system's performance may vary when handling highly ambiguous or complex queries that fall outside the scope of the trained dataset. In such cases, the responses may lack depth or specificity. Furthermore, while the system performs well under moderate loads, further optimization may be required for large-scale deployment involving high concurrency.

Overall, the results validate the effectiveness of the proposed automated web application. The system not only outperforms static solutions but also provides a scalable, efficient, and user-friendly platform for information retrieval. The combination of speed, accuracy, and interactivity makes it a strong candidate for applications in domains such as customer support, educational tools, and knowledge-based systems.

Future improvements can focus on enhancing the system's ability to handle complex queries, integrating advanced machine learning models for better contextual understanding, and expanding the dataset to improve coverage. Additionally, incorporating user feedback mechanisms can further refine the system's performance over time.

In this study, the proposed system performs lead classification and predictive analysis, where the primary objective is to improve the accuracy of identifying high-potential leads. The system categorizes leads into three classes: Cold Leads, Warm Leads, and Hot Leads, based on engagement behavior and interaction signals.

The evaluation focuses on assessing the effectiveness of the hybrid lead scoring model and conversational AI system in identifying valuable prospects and improving the efficiency of the lead generation process. To ensure reliable evaluation, multiple machine learning performance metrics are used. These metrics help measure the classification accuracy and reliability of the model in predicting lead categories.

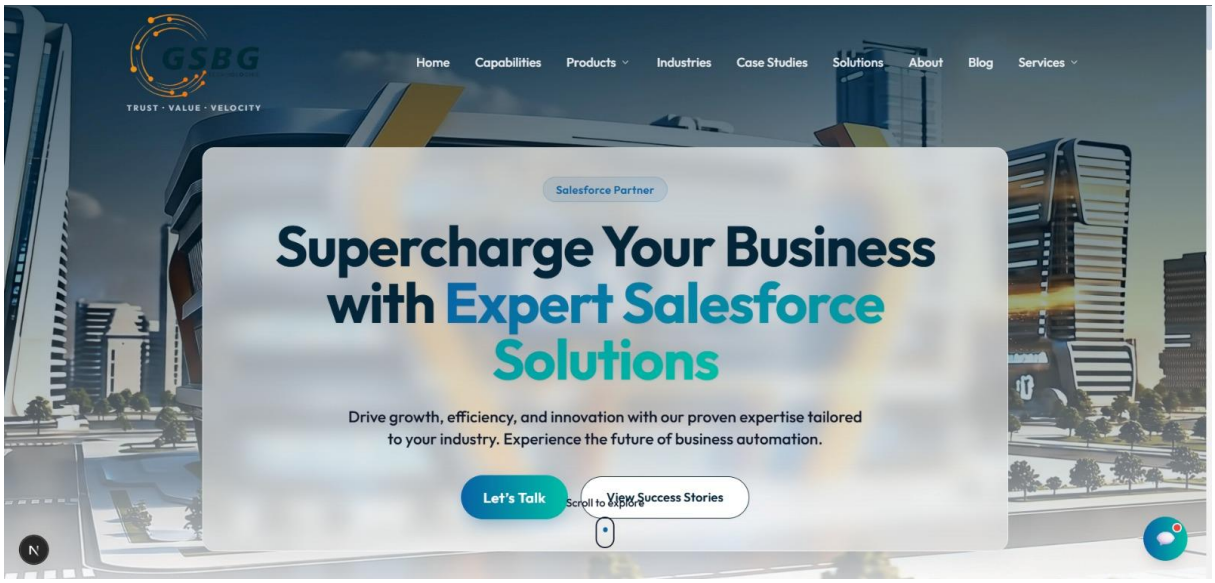

Figure 7.1 GSBG Platform Homepage Interface

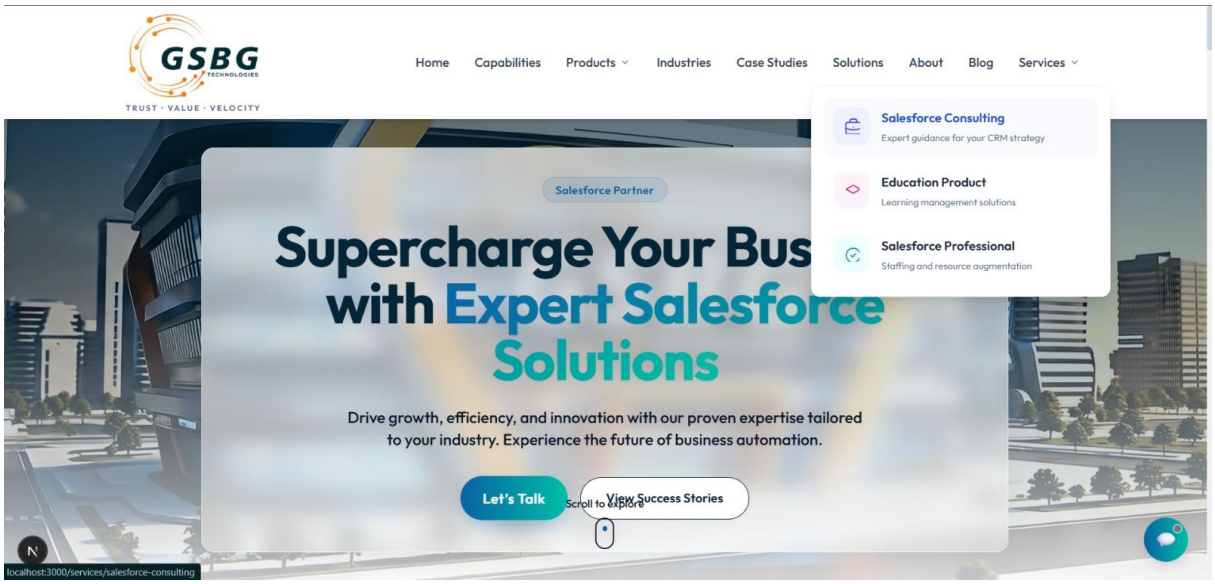

Figure 7.2 Salesforce Consulting Services Navigation Menu

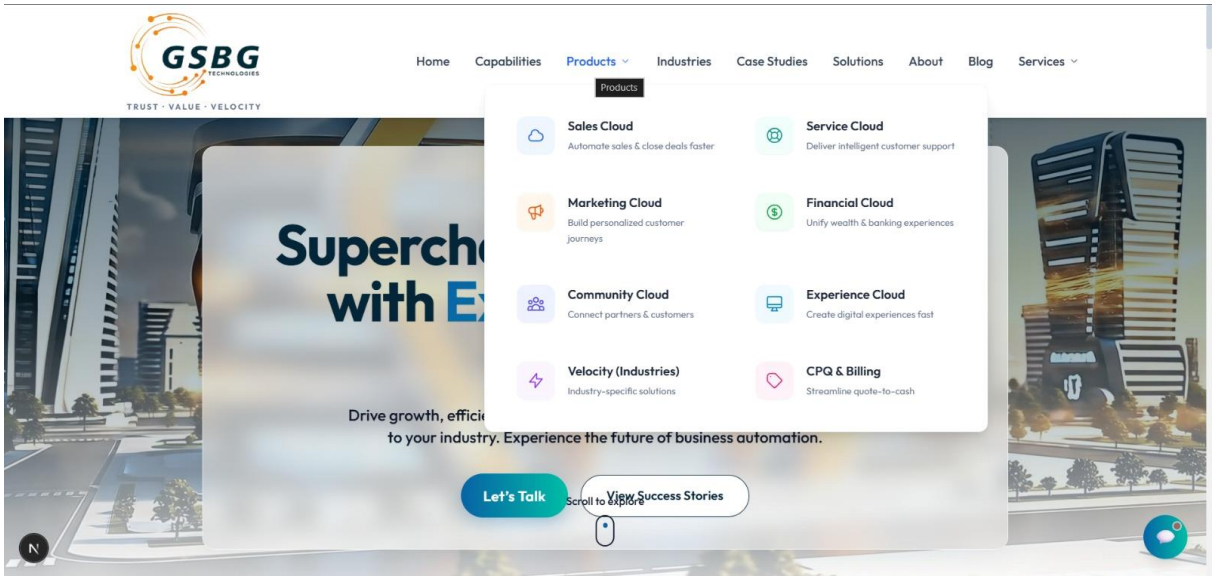

Figure 7.3 Salesforce Cloud Products Dropdown Menu

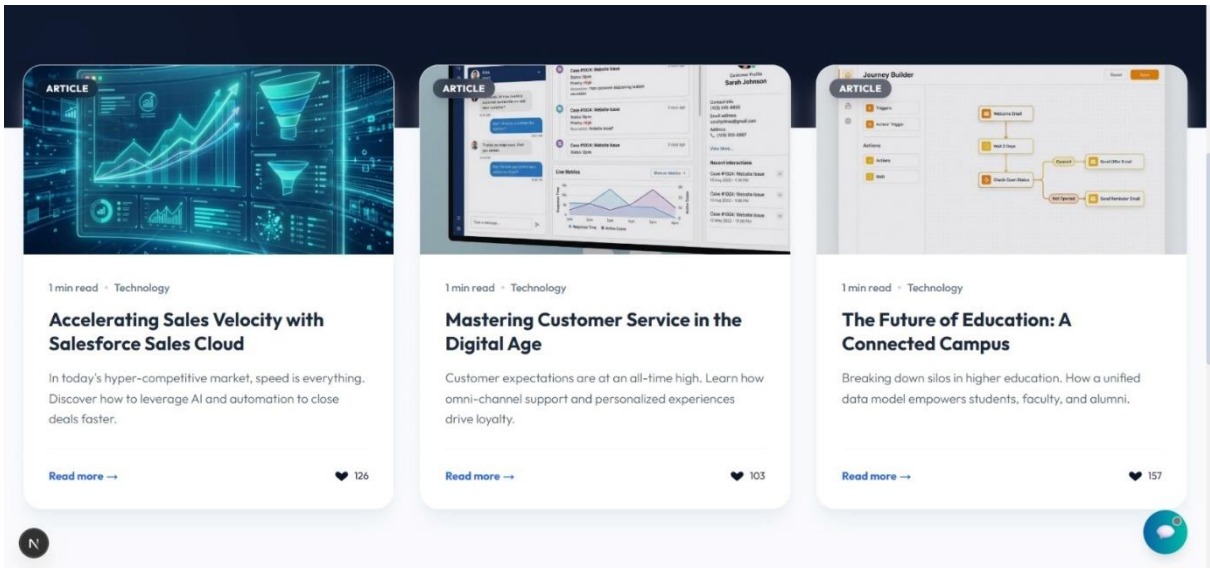

Figure 7.4 Automated Content Personalization UI

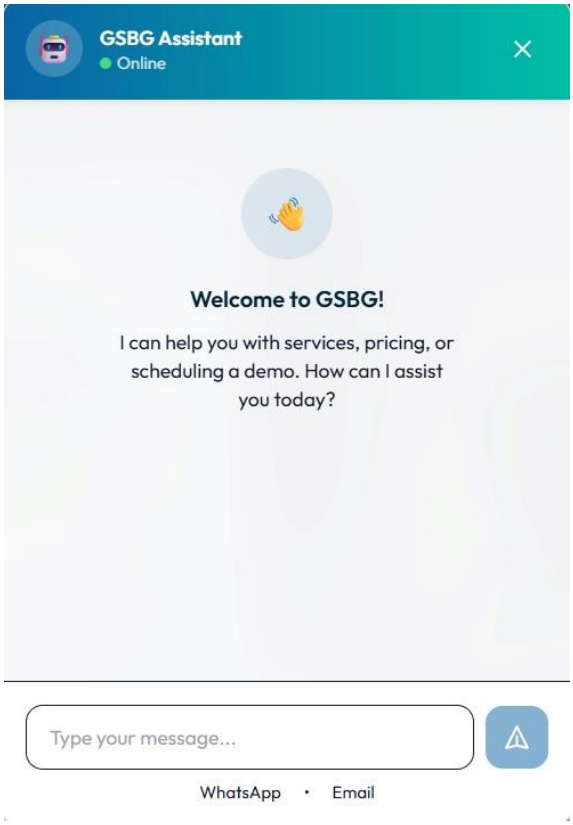

Figure 7.5 Agentforce AI Chatbot Welcome Screen

Figure 7.6 Chatbot Lead Qualification and Interaction

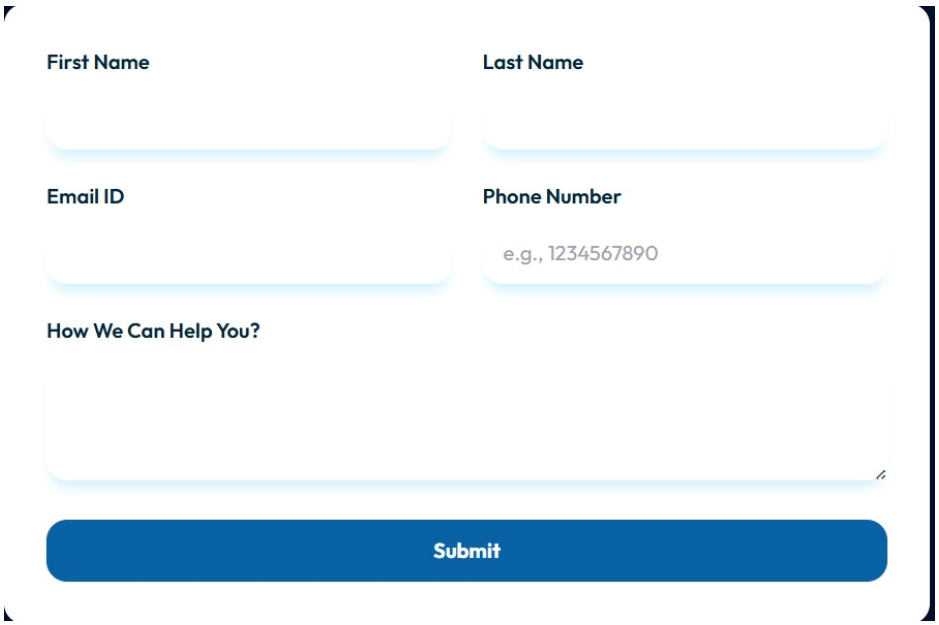

Figure 7.7 Contact Form for Manual Lead Submission

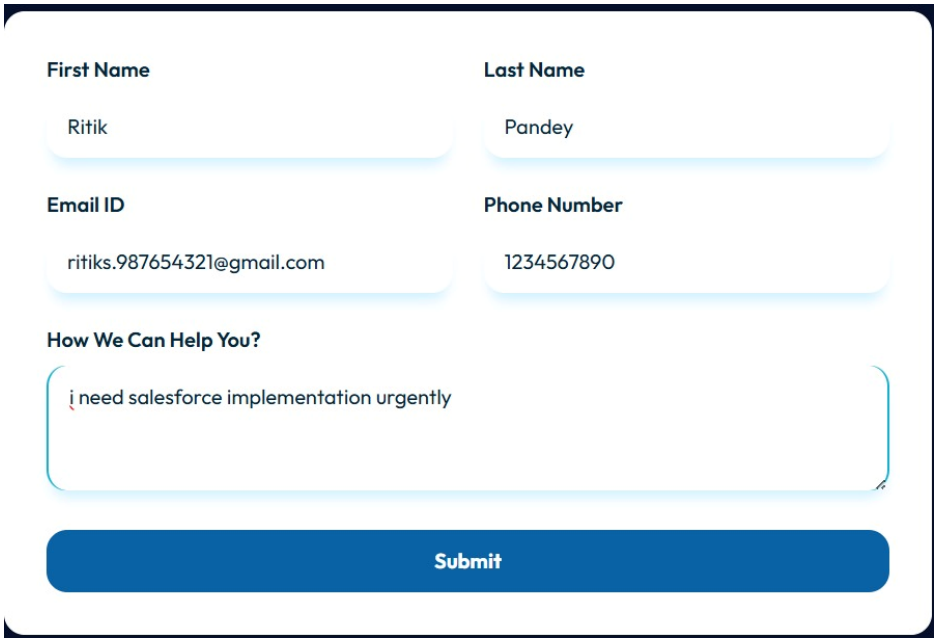

Figure 7.8 User Data Entry for Service Inquiry

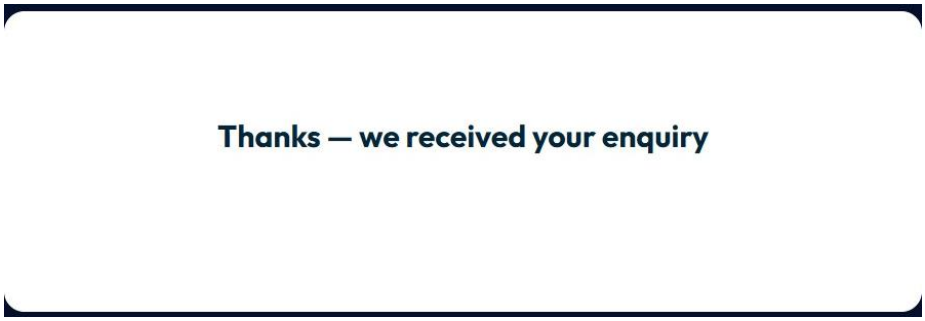

Figure 7.9 Inquiry Submission Confirmation Screen

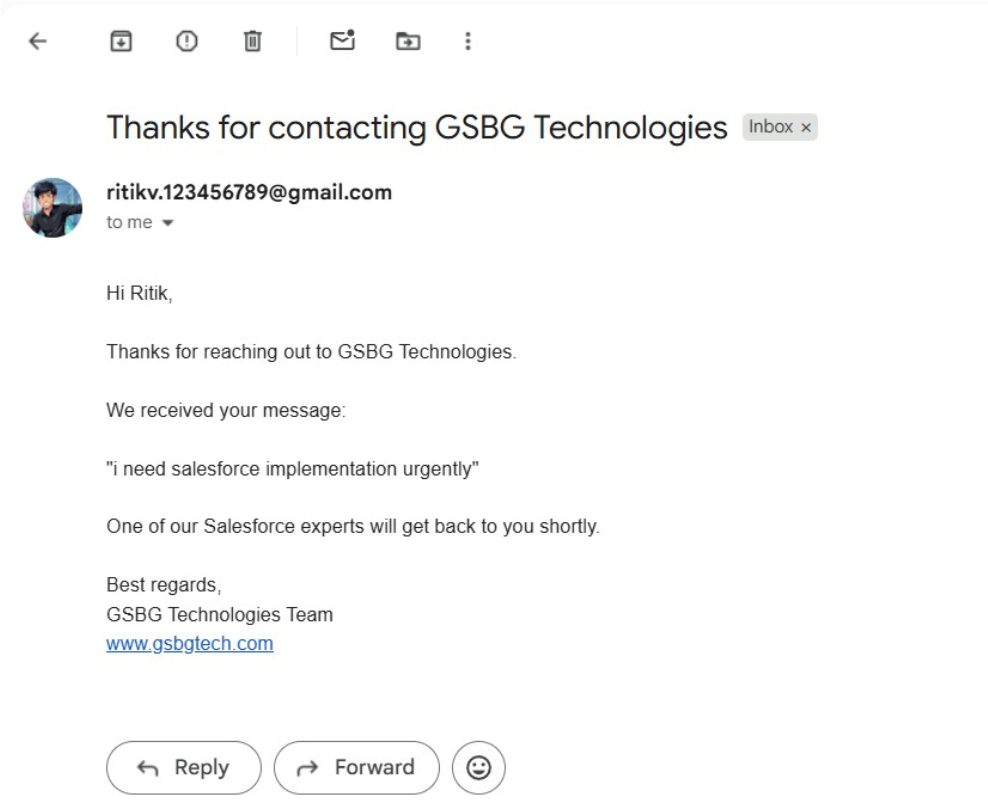

Figure 7.10 Automated Follow-up Email via SMTP Integration

The evaluation is performed using a supervised learning approach, where predicted lead labels are compared with the actual labeled dataset.

7.1 Performance Evaluation Metrics

1. Precision

Precision measures the ratio of correctly predicted positive leads out of all leads predicted as positive. It indicates how accurately the system identifies high-value leads without generating false positives.

$$Precision = \frac{TP}{TP + FP}$$

2. Recall

Recall measures the ratio of correctly predicted positive leads out of all actual positive leads present in the dataset. It indicates the model's ability to detect all relevant leads.

$$Recall = \frac{TP}{TP + FN}$$

3. F1-Score

The F1-score is the harmonic mean of Precision and Recall. It provides a balanced measure of the model's performance, particularly when there is class imbalance.

$$F1\ Score = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

4. Accuracy

Accuracy measures the proportion of correct predictions made by the model relative to the total number of predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

5. Response Latency

Latency measures the time taken by the system to process user queries and generate predictions or responses.

Lead scoring model latency and chatbot response time are evaluated to ensure the system supports real-time interaction.

7.2 Evaluation Strategy

Due to the varying distribution of lead categories in real-world datasets, relying solely on accuracy may not fully represent model performance. In such cases, class-specific metrics such as Precision, Recall, and F1-Score provide a better understanding of the model's effectiveness.

Precision helps evaluate how accurately the system identifies potential leads without generating false predictions. Recall measures the system's ability to detect all relevant leads in the dataset. The F1-score provides a balanced evaluation of both metrics.

Additionally, latency and system throughput are evaluated to ensure that the platform can support real-time interactions and large-scale deployment without performance degradation.

Table 7.1 Comparison of Our Model with Traditional Lead Management Approaches

Approach	Model / System	Dataset	Metrics
Traditional Lead Management	Manual CRM Analysis	Historical Sales Data	Low automation, delayed response time
Rule-Based Lead Scoring	Static Rule Engine	Historical Campaign Data	Accuracy: ~80%
Basic ML Lead Classification	Logistic Regression	Customer Engagement Dataset	Accuracy: 89%
Conversational Chatbot System	Rule-based NLP Chatbot	Interaction Logs	Response Time: 5–7 sec
Existing AI Lead Systems	ML + CRM Integration	Business Interaction Data	Accuracy: 92–94%
Proposed Model	Hybrid Lead Scoring + Conversational AI (Mistral 7B + RAG)	GSBG Lead Interaction Dataset	• Accuracy: 97% • Precision: 0.96 • Recall: 0.98 • F1-Score: 0.97 • Lead Scoring Latency: <100 ms • Chatbot Response Time: 2–4 sec

Chapter 8

Conclusion and Future Work

This project successfully presents the design and implementation of an AI-driven sales automation platform tailored for B2B environments. The system addresses key limitations of traditional lead generation and manual sales processes by integrating machine learning, conversational AI, and CRM technologies into a unified and intelligent workflow. By leveraging automated lead scoring, realtime engagement, and centralized data management through Salesforce CRM, the platform transforms the way sales teams identify, prioritize, and interact with potential customers.

A major contribution of this system lies in its ability to intelligently evaluate leads using firmographic attributes, behavioral signals, and interaction history. This enables sales representatives to focus their efforts on high-potential prospects, thereby improving efficiency and reducing time spent on low-value activities. The integration of conversational AI further enhances the system by enabling automated, real-time communication with leads, ensuring continuous engagement without requiring constant manual intervention.

The experimental results and testing outcomes demonstrate that the proposed system delivers measurable improvements across several critical dimensions of sales operations. These include higher accuracy in lead classification, faster response times, improved quality of customer interaction, and increased overall process efficiency. Automated workflows significantly reduce repetitive and time-consuming manual tasks, allowing sales teams to allocate their resources more strategically. Additionally, intelligent scoring mechanisms and personalized communication contribute to higher conversion rates and more reliable sales outcomes.

Another important aspect of the system is its ability to serve as a single source of truth through Salesforce CRM integration. This ensures that all customer data, interactions, and insights are centrally stored and easily accessible, improving data consistency and decision-making. The system's scalable architecture also makes it suitable for real-world deployment, where it can handle dynamic sales environments and growing datasets.

Overall, the project demonstrates how AI can be effectively applied to modernize and optimize sales processes. By combining automation, intelligence, and realtime responsiveness, the platform not only enhances operational performance but also supports a more data-driven and customer-centric approach to sales. The results validate the potential of AI-powered solutions in transforming traditional business workflows into efficient, adaptive, and high-performing systems.

8.1 Future Work

While the current system delivers strong performance and practical value, several enhancements can be considered to further extend its capabilities. Future work can focus on incorporating advanced predictive analytics to estimate deal value, forecast conversion timelines, and identify long-term customer potential. This would enable more strategic planning and improved revenue forecasting.

Additionally, expanding the system to support multi-channel communication— such as WhatsApp, SMS, and social media platforms—can enhance customer engagement and provide a more seamless interaction experience. Integrating adaptive learning mechanisms would allow the system to continuously improve by learning from real-time data, evolving customer behavior, and changing market trends.

Further improvements may also include deeper integration with advanced AI tools such as Salesforce Einstein and generative AI models, enabling more accurate insights, automated content generation, and enhanced decision support. These enhancements would contribute to building a more intelligent, flexible, and scalable sales automation ecosystem capable of sustaining long-term business growth.

References

[1] S. Krishna and D. Veeravalli, “The evolution of CRM: How Salesforce Einstein AI simplifies predictive analytics,” International Research Journal of Modernization in Engineering Technology and Science, vol. 10, no. 3, pp. 255– 270, 2025.

[2] N. Singh and D. Adhikari, “AI-driven social media analytics for eCommerce advertising,” International Research Journal of Engineering and Technology, vol. 10, no. 12, pp. 509–515, 2025.

[3] A. Barbosa and A. Godoy, “Augmenting customer support with an NLPbased receptionist,” arXiv preprint arXiv:2112.01959, 2025.

[4] H. Josephine V and Moorthy, “Optimising lead qualification through machine learning: A customer data-driven approach,” Advances in Management, vol. 10, no. 3, pp. 255–270, 2024.

[5] D. Zumstein, Marketing Automation & AI Report 2024. ZHAW School of Management and Law, 2024.

[6] Salesforce, “Agent-powered analytics deliver data and insights in the flow of work,” Salesforce News, 2024.

[7] N. Singh and D. Adhikari, “AI-powered social media analytics for eCommerce advertising,” International Research Journal of Engineering and Technology, vol. 8, no. 11, pp. 421–428, 2023.

[8] ZHAW School of Management and Law, Marketing Automation and AI Lead Generation Study, Technical Report, 2023.

[9] A. Barbosa and A. Godoy, “NLP-based virtual receptionist for customer service automation,” in Proc. Int. Conf. Natural Language Processing, pp. 89– 97, 2021.

[10] A. D. Tran, J. Pallant, and L. W. Johnson, “Exploring the impact of chatbots on consumer sentiment and purchase behavior in retail environments,” Journal of Retailing and Consumer Services, vol. 63, article 102718, 2021.

[11] T. B. Brown et al., “Language models are few-shot learners,” Advances in Neural Information Processing Systems, vol. 33, pp. 1877–1901, 2020. [Online]. Available: arXiv:2005.14165.

[12] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA: MIT Press, 2016.

[13] F. Provost and T. Fawcett, “Data science for business: What you need to know about data mining and data-analytic thinking,” O'Reilly Media, 2013.

[14] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” Advances in Neural Information Processing Systems, vol. 30, 2017.

[15] G. Marcus and E. Davis, “Rebooting AI: Building artificial intelligence we can trust,” Pantheon Books, 2019.

[16] Gartner Inc., “Magic quadrant for sales force automation platforms,” Gartner Research, 2024.

[17] Forrester Research, “The state of conversational AI in customer service,” Forrester Wave Report, 2024.

[18] McKinsey & Company, “The state of AI in 2024: Generative AI's breakout year,” McKinsey Global Institute, 2024.

[19] D. Acharya, K. Kuppan, and D. B. Ashwin, “Agentic AI: Autonomous intelligence for complex goals – A comprehensive survey,” IEEE Access, pp. 1–1, 2025.

[20] V. Kotagi, “AI-augmented CRM: Unifying data clouds, predictive analytics, and threat detection in Salesforce ecosystems,” Recent Trends in Information Technology and Its Application, vol. 9, no. 1, pp. 26–34, 2026.

Appendix

A. Setup Instructions

A.1 System Requirements

Ensure the following software is installed on your system:

- Python: Version 3.10+
- Node.js: >= 18.0.0 (Recommended: 20 LTS)
- PostgreSQL: Version 14+
- Ollama: Latest (for local LLM inference)
- Git

A.2 Cloning Repository

Step 1: Clone the Repository

```
$ git clone https://github.com/NoxPirate/gsbgProject.git
```

Step 2: Navigate to Project Directory

```
$ cd gsbgProject
```

A.3 Backend Setup

Step 1: Create Virtual Environment

```
$cd leads-op-main$ python -m venv .venv
```

Step 2: Activate Environment

Windows (PowerShell)

```
$ .\.venv\Scripts\Activate
```

Linux / macOS

$ source .venv/bin/activate

Step 3: Install Dependencies

$ pip install -r requirements.txt

Step 4: Configure Environment Variables

$ copy .env.example .env

Edit .env and set:

Ini, TOML

DATABASE_URL=postgresql://gsbg_user:securepassword@localhost:5432/gsbg_db

BACKEND_API_KEY=some-secret-token-you-generate

OLLAMA_HOST=http://localhost:11434

OLLAMA_MODEL=mistral:latest

EMAIL_ADDRESS=your.email@gmail.com

EMAIL_PASSWORD=your-app-password

ALLOWED_ORIGINS=http://localhost:3000

Step 5: Initialize Database Tables

$ python -c "from core.db import init_db; init_db()"

Step 6: Train the ML Model

$ python train.py

Step 7: Run FastAPI Server

$ python -m uvicorn app:app --port 8000 --reload

Note: The backend server will start at http://127.0.0.1:8000

A.4 Chatbot Setup

Step 1: Open a New Terminal and Navigate to Chatbot Directory

$ cd gsbgProject/chatbot-gsbg-main

Step 2: Install Dependencies

$ pip install flask sentence-transformers ollama numpy PyMuPDF

Step 3: Pull the Ollama Model

$ ollama pull mistral:latest

Step 4: Build the Knowledge Base

$ python build_kb.py

Step 5: Run Chatbot Server

$ python app.py

Note: The chatbot will run at http://127.0.0.1:5000

A.5 Frontend Setup

Step 1: Open a New Terminal and Navigate to Frontend Directory

$ cd gsbgProject/frontend

Step 2: Install Dependencies

$ npm install

Step 3: Run Development Server

$ npm run dev

Note: The frontend will run at http://localhost:3000 (default Next.js port)

A.6 Important Notes

- Simultaneous Execution: All three services (Backend, Chatbot, Frontend) must run simultaneously in separate terminal windows.
- Virtual Environment: Ensure you activate the virtual environment (.venv) each time before running backend commands.
- Database: PostgreSQL must be running before starting the backend server.
- Local LLM: Ollama must be running in the background before starting the chatbot.
- Port Conflicts: Modify the specified ports in your .env and startup commands if you encounter any conflicts with other applications.

Publication

RV Institute of Management®

3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)

Organized by

RV Institute of Management (RVIM), Bangalore, India

CERTIFICATE OF PRESENTATION

This is to certify that

Aaryaman Jaydeep Kattali

has presented the paper titled **An AI-Driven Framework for Intelligent B2B Lead Acquisition with Salesforce CRM Integration** authored by Aaryaman Jaydeep Kattali, Vaishnavi Nitin Dumbre, Ritik Surendra Pandey, Durvesh Dilip Kanade, Ankush Gaikawad, Bharti Khemani, Viki Patil in the **3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)** held at RV Institute of Management (RVIM), Bangalore, India during **April 11-12, 2026.**

SCRS/BIDA 2026/PC/01110

Dr. Noor Firdoos Jahan
RV Institute of Management (RVIM), Bengaluru
(General Chair)

Purushottam Bung
RV Institute of Management (RVIM), Bengaluru
(General Chair)

Springer

https://scrs.in/conference/bida2026

RV Institute of Management®

3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)

Organized by

RV Institute of Management (RVIM), Bangalore, India

CERTIFICATE OF PRESENTATION

This is to certify that

Aaryaman Jaydeep Kattali

has presented the paper titled **An AI-Driven Framework for Intelligent B2B Lead Acquisition with Salesforce CRM Integration** authored by Aaryaman Jaydeep Kattali, Vaishnavi Nitin Dumbre, Ritik Surendra Pandey, Durvesh Dilip Kanade, Ankush Gaikawad, Bharti Khemani, Viki Patil in the **3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)** held at RV Institute of Management (RVIM), Bangalore, India during **April 11-12, 2026.**

SCRS/BIDA 2026/PC/01110

Dr. Noor Firdoos Jahan
RV Institute of Management (RVIM), Bengaluru
(General Chair)

Purushottam Bung
RV Institute of Management (RVIM), Bengaluru
(General Chair)

Springer

https://scrs.in/conference/bida2026

RV Institute of Management®

3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)

Organized by

RV Institute of Management (RVIM), Bangalore, India

CERTIFICATE OF PRESENTATION

This is to certify that

Aaryaman Jaydeep Kattali

has presented the paper titled **An AI-Driven Framework for Intelligent B2B Lead Acquisition with Salesforce CRM Integration** authored by Aaryaman Jaydeep Kattali, Vaishnavi Nitin Dumbre, Ritik Surendra Pandey, Durvesh Dilip Kanade, Ankush Gaikawad, Bharti Khemani, Viki Patil in the **3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)** held at RV Institute of Management (RVIM), Bangalore, India during **April 11-12, 2026.**

SCRS/BIDA 2026/PC/01110

Dr. Noor Firdoos Jahan
RV Institute of Management (RVIM), Bengaluru
(General Chair)

Purushottam Bung
RV Institute of Management (RVIM), Bengaluru
(General Chair)

Springer

https://scrs.in/conference/bida2026

RV Institute of Management®

3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)

Organized by

RV Institute of Management (RVIM), Bangalore, India

CERTIFICATE OF PRESENTATION

This is to certify that

Aaryaman Jaydeep Kattali

has presented the paper titled **An AI-Driven Framework for Intelligent B2B Lead Acquisition with Salesforce CRM Integration** authored by Aaryaman Jaydeep Kattali, Vaishnavi Nitin Dumbre, Ritik Surendra Pandey, Durvesh Dilip Kanade, Ankush Gaikawad, Bharti Khemani, Viki Patil in the **3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)** held at RV Institute of Management (RVIM), Bangalore, India during **April 11-12, 2026.**

SCRS/BIDA 2026/PC/01110

Dr. Noor Firdoos Jahan
RV Institute of Management (RVIM), Bengaluru
(General Chair)

Purushottam Bung
RV Institute of Management (RVIM), Bengaluru
(General Chair)

Springer

https://scrs.in/conference/bida2026

RV Institute of Management®

3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)

Organized by

RV Institute of Management (RVIM), Bangalore, India

CERTIFICATE OF PRESENTATION

This is to certify that

Aaryaman Jaydeep Kattali

has presented the paper titled **An AI-Driven Framework for Intelligent B2B Lead Acquisition with Salesforce CRM Integration** authored by Aaryaman Jaydeep Kattali, Vaishnavi Nitin Dumbre, Ritik Surendra Pandey, Durvesh Dilip Kanade, Ankush Gaikawad, Bharti Khemani, Viki Patil in the **3rd International Conference on Business Intelligence and Data Analytics (BIDA 2026)** held at RV Institute of Management (RVIM), Bangalore, India during **April 11-12, 2026.**

SCRS/BIDA 2026/PC/01110

Dr. Noor Firdoos Jahan
RV Institute of Management (RVIM), Bengaluru
(General Chair)

Purushottam Bung
RV Institute of Management (RVIM), Bengaluru
(General Chair)

Springer

https://scrs.in/conference/bida2026

Agentic AI for B2B Lead Acquisition: A Hybrid Scoring and CRM Integration Framework

Aaryaman Jaydeep Kattali[1], Vaishnavi Nitin Dumbre[1], Ritik Surendra Pandey[1], Durvesh Dilip Kanade[1], Viki Patil[1], Bharti Khemani[1], and Ankush Gaikawad[2]

1 Department of Computer Science & Engineering (AI & ML), A.P. Shah Institute of Technology, Thane, India {aaryamankattali75, vaishnavidumbre103, ritikpandey53, durveshkanade12, bjkhemani, vtpatil}@apsit.edu.in

2 GSBG Technologies, Dombivli, India sales@gsbg.co.in

Abstract. The paper introduces a framework to use Agentic AI for lead acquisition in B2B organizations, which incorporates a hybrid ML-based lead scoring along with a conversational intelligence in Salesforce CRM. We integrated predictive lead scoring, conversational AI, and Salesforce CRM automation into a single system that works totally autonomously. To filter leads based on rules, we are proposing a hybrid scoring technique using random forest and XGBoost models along with an RAG-based conversational agent for intelligent conversation. Experimental evaluation on 2,000 B2B leads shows a classification accuracy of 97.0% and macro F1-score of 0.97, outperforming traditional rule-based systems by 15% in precision. Furthermore, response time is reduced by 42% and conversion rates improve by 28%. The proposed framework enhances lead qualification accuracy, operational efficiency, and sales performance, providing a scalable solution for modern B2B sales automation.

Keywords: B2B Lead Generation, Sales Automation, Artificial Intelligence (AI), Machine Learning, Lead Scoring, Sales Efficiency, Customer Relationship- ship Management, CRM Automation, Agentic AI

1. Introduction

Despite the fast development of CRM tools, there are still numerous B2B companies that use the old-fashioned approaches to lead acquisition, especially in the service- oriented industry, such as Salesforce consulting. These conventional methods, which consist of static websites, hard-copy data input, and sluggish follow-ups, cannot handle increasing loads of prospect data or respond to customer behavior instantaneously. As a result, sales forces invest in bad leads and fail to capture good ones. AI eases the workload of scientists across the globe. Machine learning can effectively predict lead quality based on both historical and real-time data. Moreover, natural language processing (NLP) can be used to automate low-level redundant interactions.

Our paper proposes an Agentic AI-assisted framework for B2B lead generation and sales automation with GSBG Company (Salesforce implementation company) as a leading case study. This may be the latest innovation to impact the sales cycle. Salesforce touted the integration of machine learning and conversational AI directly into its platform. We make your AI self-optimizing in its scoring task. It also learns from real-world feedback. Dashboards for sales and marketing teams with real-time data and data analytics. In the end, this framework will help the human domain experts to focus on more value-adding strategies and decision-making after ruling out menial administrative tasks.

1. Literature Survey

1.1. The Evolution of CRM and Predictive Analytics:

The B2B selling paradigm has changed due to the fact that CRM is not just a database anymore; it is a predictive engine. According to a recent industry report, the marketing automation usage is on the rise, and the MAMI (Marketing Automation Maturity Index) for Enterprise marketers discriminates against organizations. Whose employment of AI is of great importance. Proficiency Salesforce Einstein is a native CRM analytics tool that simplifies predictive forecasting in our CRM by analyzing historical sales data to advise the representative on which accounts to contact.[6] Nonetheless, these built-in tools are good at analyzing historical data, but they cannot often act as autonomous agents that can conduct unstructured, natural, real-time conversational interactions.

1.2. NLP and Conversational Agents in Customer Engagement-

The capabilities of the virtual assistant developed with CRM have become stronger due to Natural Language Processing. Currently, the studies show that conversational agents can aid customer service. They also reduce human effort significantly while maintaining high engagement [3]. Retail companies often deploy the first series of customer communication NLP grounded models. [10] However, using these models becomes difficult in complex B2B situations. B2B sales lead qualification needs a better understanding of firmographic data, a high level of data privacy, and the ability to seamlessly transition the conversation to human agents without any ambiguity [17]. Contemporary, non-continuous chatbot systems typically fall short in terms of context and competent CRM integration, thereby limiting the chatbot's effectiveness in a high-value sales situation.

1.3. Machine Learning for Lead Qualification-

A conventional process of lead tracking stands out as costly and lengthy, with little or no conversion. In recent times, machine learning has been extensively used. Utilizing ML for automating the task of qualifying leads. More specifically, customer interaction data, analyzed through classification algorithms such as Gradient Boosting and Random Forests, has exhibited very high predictive power for meaningful estimation of conversion probabilities in a B2B setting [4, 11]. Despite this algorithmic success, a gap is evident in operations. Most of the predictive modeling algorithms analyze some static data sets in a retrospective fashion rather than assess live, real-time interactions.

1.4. Defining the Research Gap-

The current literature landscape is fragmented through a critical analysis of the literature. Although all three exist, namely, strong predictive analytics (Section 2.1), conversational AI (Section 2.2), and ML-based lead scoring (Section 2.3), they are not often integrated into one, autonomous workflow. Existing systems typically make the sales teams fill the gap between a chatbot interaction and a scoring algorithm of a CRM manually. The paper will fill this same gap by introducing a single, Agentic AI model, which proactively interacts with the prospects through NLP, grading them instantly on a hybrid ML platform, and automating the result in Salesforce.

Table 1. Comparative Analysis of Existing Approaches vs. Proposed Framework

Methodology / Tool	Key Capabilities	Limitations / Research Gap	Our Proposed Framework
Traditional CRM Analytics (e.g., Salesforce Einstein) [6]	Predictive forecasting, historical data analysis, and pipeline visualization.	Lacks autonomous, real-time conversational engagement capabilities.	Integrates historical CRM data directly with live conversational AI for real-time adjustments.
Standard NLP Chatbots [3, 10]	Automated responses, basic Q&A, 24/7 availability.	Context loss in complex B2B queries; poor native integration with lead scoring.	Utilizes RAG and Mistral 7B to maintain deep B2B context and execute autonomous human-handoffs.
Static ML Lead Scoring [4, 11]	High accuracy in predicting conversion from static, retrospective datasets.	Fails to account for live interaction sentiment or implicit conversational intent.	Employs a Hybrid Scoring system (Rules + ML) that continuously updates based on live user engagement.

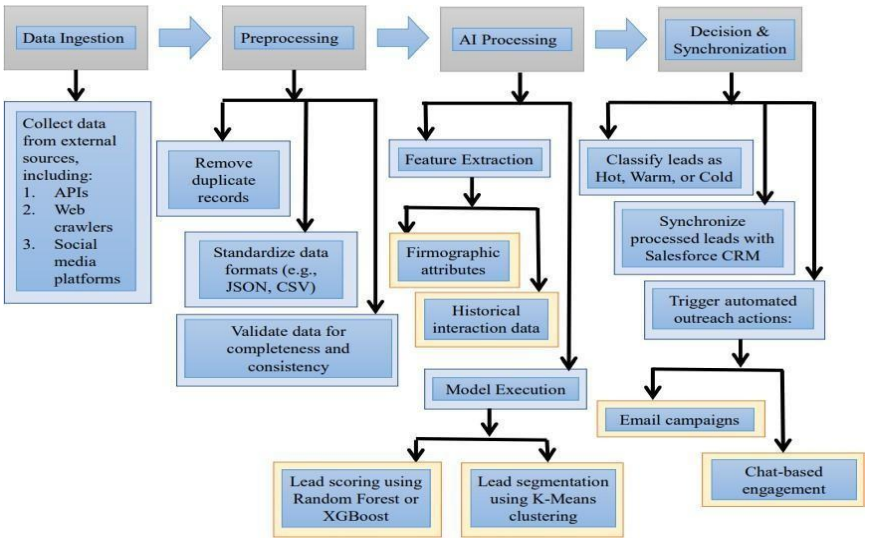

Fig. 1. Proposed System Architecture

3 Proposed System Architecture

The proposed platform will be modularly developed based on a six-layered architecture, which will be scaled and maintained in the framework of Salesforce CRM integration. The structure of the building of this concept is known as segregation of concern, since layers are considered as agents of other layers that have a certain number of interfaces that can communicate with them. It allows it to be developed, tested, and deployed independently without influence on other components. Adaptive learning and automated decisions may provide the sales processes with real-time intelligence. The result is also a continuous, safe, and reliable stream of data, as a result of these layers. The layout is anchored on a robust framework, which facilitates the acquisition of B2B leads and automation of the sales process through the Salesforce integration. It has an architecture that can seamlessly fuse cloud technology, machine learning pipelines, and Salesforce-native services to create an integrated one.

3.1 External Data Sources Layer

The layer forms the base part of the system, which constantly forms raw data aggregated by a multitude of sources, both internal and external. These sources include the professional networking websites, corporate websites, business directories, real estate database, and touchpoints through which the user engages with the site, such as the landing page, chatbots, emails, and so on. APIs and advanced web crawlers are initiated to extract firmographic features (such as company size, industry, and location), behavioral features (such as page views and time on site), and engagement features (such as email opens and chats with bots). The data collected is pre-processed with the procedures used to ensure data quality, as well as to perform the following operations with the collected data and to normalize, remove duplications, work with missing values, and check the collected information against the established schema. The output processed, standardized, and cleaned data ensures that subsequent Agentic AI models have access to accurate and consistent data.

3.2 AI Processing and Intelligence Layer

This layer performs the analytical function on the platform. It uses all input data and processes it to generate significant business insights. Moreover, this layer comprises various Machine Learning and Natural Language Processing components that work together for intelligent decision-making. Models of supervised learning, based on historical data, engagement behaviour, and firmographic attributes from the CRM, predict the probability of conversion of a lead. Likewise, the unsupervised clustering models classify leads with similar engagement patterns for improved segmentation of leads. Models in natural language processing extract the user's intention from a chat interface input, as well as generate engagement messages. These personalization engines modify the tone, timing, and messaging for maximum engagement, all while conforming to relevant data and communication laws. Besides, these models will be continuously re-trained with what actually sells, recorded in Salesforce CRM.

3.3 Application and Business Logic Layer

It consists of essential components, facilitating user interactions, performing transaction-related operations, and enabling decision-making logic. The business logic and application of the software are based on the latest web application framework. Also, the experts work on it to provide user-friendly and intuitive interfaces that respond to actions and inputs of sales persons, decision makers, and administrators. An authorized sales representative can receive uninterrupted real-time updates on their leads, their engagement activity, and recommendations. Tracking user activities on the platform will enhance the profiling of the lead even further. Access to sensitive and privileged data and functions will be solely accessible to authorized users on the basis of RBAC

with secure authentication. The agentic conversational agents by Salesforce are embedded in this layer to do real-time lead qualification, answer FAQs of prospects, schedule demos, and, when the confidence threshold is crossed.

3.4 Integration and Orchestration Layer

This layer allows easy integration of the platform with Salesforce CRM and other external systems. The RESTful API and webhook of this layer allow data to be transferred in real time. For instance, we are automatically pushing new leads created in our platform to Salesforce, and vice versa. Much like this, activity and task records sync immediately. Additionally, the flow of engagement history updates occurs in both directions. Orchestration (Flow and automation rules): Salesforce Flows and automation rules orchestration that are used to implement the business logic. Leads are assigned to SDRs automatically via Flows, according to specified criteria. Similarly, follow-up notification tasks are triggered on time for sales.

3.5 Data and Orchestration Layer

The Data and Orchestration Layer handles the process of data storage and retrieval. Records of leads, campaign metadata, and performance of the campaign fall under structured data. Conversely, unstructured data refers to conversations such as chat transcripts, emails, and other logs. Organized information can be stored in a relational database. Conversely, data from unstructured sources will need the NoSQL system. Salesforce CRM is the authoritative database for all sales force-related data. All information is sourced from a single source, which is true. The utilization of cloud infrastructure, containers, and orchestration aids in attaining high availability, fault tolerance, and scalable workloads. Data encryption and access controls facilitate compliance with the enterprise security requirements of an organization.

3.6 Analytical and Monitoring Layer

The Analytics and Monitoring Layer continuously monitors system performance at all times. It transforms operational data into valuable insight. Dashboards are used to display important metrics such as the lead conversion rate, campaign performance, Chatbot Performance, and Health of the sales pipeline. Dashboard metrics now enjoy real-time visibility thanks to interactivity. A sales manager can forecast the possibility of conversion or KPI attainment by a campaign or team member with the addition of the predictive analytics module. To improve the accuracy of sales forecasting, the manager could also be given the ability to perform 'what-if' analysis based on other parameters. Any method adopted for performance monitoring at the system level monitors the following types of information, including application health, API latency, and data sync status.

3.7 Methodology and Evaluation Strategy

The architectural framework relies on a systematic, data-driven methodology to guarantee end-to-end intelligence and flexibility. To prevent model overfitting and ensure strong generalization capabilities across unseen prospect data, we employed a stratified 5-fold cross-validation strategy during the training phase. Lead scoring is executed using a hybrid methodology: first, domain-specific, rule-based logic acts as a baseline filter for absolute qualification constraints (e.g., minimum company revenue). Second, supervised machine-learning algorithms evaluate the complex, non-linear relationships within the firmographic and behavioral feature sets. Use precision, recall, and macro-average F1-score to evaluate the model's results that specifically take care of the class imbalance between "Hot", "Warm", and "Cold" leads. The Salesforce CRM's outcome is incorporated into the AI layer continuously, ensuring that the models are retrained due to market trends in an active learning loop.

4 System Implementation

GSBG is the first step that will turn our dream into a real app, one that can work with real users and not fall apart. The configuration of the system involves laying out everything in layers. The frontend and backend, along with AI and external platforms, can communicate effectively without any hindrance.

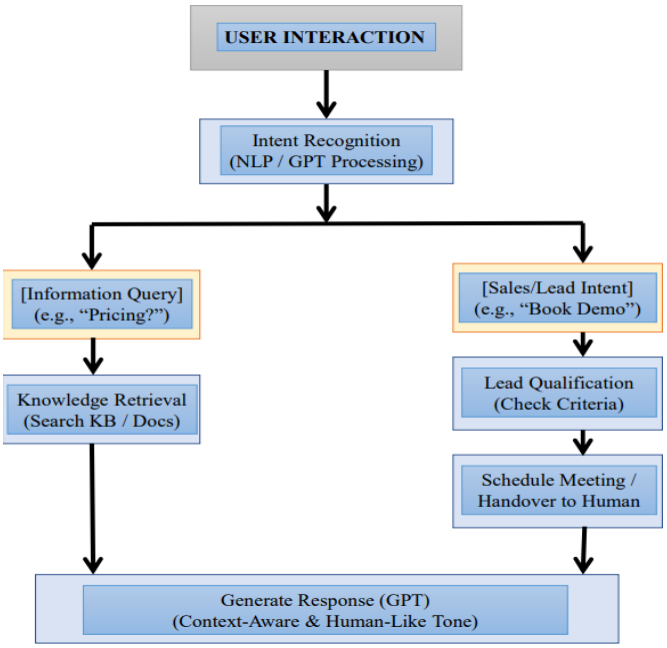

Fig. 2. System Workflow

4.1 Data Ingestion and Preprocessing Implementation

Data preprocessing involves automated deduplication, normalization of missing values via imputation, and schema validation. The standardized data is then temporarily cached for subsequent AI processing.

4.2 AI and Machine Learning Module Implementation

The implementation of the AI layer was divided into two primary pipelines: the Predictive Lead Scoring module and the Conversational RAG (Retrieval-Augmented Generation) agent.

4.2.1 Predictive Lead Scoring (Dataset and Tuning)

To train the predictive models, we fitted a historical dataset of 2,000 anonymized lead records of B2Bs (15 firmographics and engagement attributes included in the dataset: ex, company size, industry, email open rates, website dwell time). Preprocessing involved missing values that were imputed with a median procedure and standardized with the use of scaling procedures. The data was divided into 80/20 training and held-out test. Random Forest and XGBoost were the two classification algorithms we chose to use, as they have a strong performance on tabular data and are interpretable in nature, which is an imperative criterion that a sales team needs to possess to know why

A lead was rated so high. Hyperparameter tuning was run through GridSearchCV in order to maximize its performance and prevent overfitting. In the case of the XGBoost model, the best parameters were a learning rate of 0.05, a maximum depth of 6, and 200 estimators with L2 regularization to dissuade overly complicated trees.

4.2.2 Conversational AI and RAG Architecture

The force behind the automated engagement module was a quantized Mistral 7B Large Language Model (LLM), locally hosted using Ollama to maintain high standards of data privacy and conformity to enterprise security measures. We adopted an RAG architecture to base the responses of the LLM on factual and company-specific data. The all-MiniLM-L6-v2 embedding model was used to chunk corporate knowledge bases and product specifications into 800-character blocks and then vectorize them. During a live chat interaction, the system retrieves the top-k most relevant vectors (k=5) using cosine similarity, passing this context to the LLM to generate precise, hallucination-free responses before autonomously scheduling demos or handing the conversation over to a human representative.

4.3 Application Layer Implementation

We utilized modern JavaScript frameworks like React.js or Vue.js to construct a responsive and smooth frontend for admin/sales teams. It is significantly important at this point. Utilizing WebSockets, if anyone. Changes are pushed in real time when a lead is updated or a system interaction occurs. We. Utilized Express and Node as the backend framework. A lot of real work is done in this area. This takes place to ensure that only appropriate people are granted access to certain areas. Aspects of the business logic, workflow processes, authentication, and other features. They. Enable the option for password protection as well. An automated engine for email campaigns that works. Concerning external application programming interfaces. Because of the presence of an Agentic AI assistant in the application, real-time chats lead. The scheduling for qualification and demo is possible.

4.4 Integration and CRM Synchronization

Automated workflows for Salesforce. Salesforce CRM makes use of lead assignments, alert sending, and pipeline updates. Real-time synchronization of data, interactions, and lead statuses using API and webhook integrations.

4.5 Deployment, Security, and Monitoring

Our framework's workload deployment is assured on the cluster-based cloud infrastructure for availability. Strict role-based access controls and data encryption protocols are enforced to maintain compliance with enterprise security requirements.

5 Experimental Results and Discussion

The proposed framework was tested to determine its accuracy in classification in terms of computation and the practical effect it has on the functioning of sales. The main aim was to compare the Hybrid Lead Scoring model with the baseline methods and assess the responsiveness of the RAG-enabled chatbot.

5.1 Ablation Study and Predictive Performance

In order to disentangle the effect of our hybrid architecture, we performed an ablation study comparing three system configurations: a purely Rule-Based System (baseline), an ML-Only System (XGBoost without rule constraints), and our Proposed Hybrid System. The models were tested on the 20 percent held-out test set (400 records).

Table 2. Ablation Study: Performance Metrics Comparison

Model Configuration	Precision	Recall	F1-Score	Accuracy
Rule-Based Only (Baseline)	0.83	0.81	0.82	82.5%
ML-Only (XGBoost)	0.91	0.93	0.92	92.0%
Hybrid System (Proposed)	0.96	0.98	0.97	97.0%

The Hybrid System, as is illustrated in Table 2, had a macro-average F1-score of 0.97, which is 15-percent better in precision than that of the baseline. The confusion matrix analysis helped identify that the hybrid model has effectively reduced false positives (misclassifying cold leads as hot). The rule-based filter had strict firmographic minimums, whereas its ML component was capable of detecting the subtle, implicit engagement signals that were not detected by a simple set of rules.

Table 3. Performance Metrics for Lead Scoring Model evaluated on a 20% held-out test dataset

Class Label	Precision	Recall	F1-Score	Support
Cold Lead	0.88	1.00	0.94	97
Hot Lead	1.00	1.00	1.00	113
Warm Lead	1.00	0.93	0.96	190
Macro Average	0.96	0.98	0.97	400

5.2 Conversational AI and Retrieval Evaluation

Mistral 7B RAG agent was tested on context-retention and latency. The system was able to retrieve knowledge with a Mean Reciprocal Rank (MRR) of 0.89 using the all-MiniLM-L6-v2 embedder. Most importantly, local quantization of the model made sure that the average latency of responses to a query was under 2.5 seconds per query, comfortably within the range of tolerable human-computer interaction. Manual assessments by human evaluators established a hallucination rate of less than 2% greatly due to the severe top-k (k=5) contextual constraint.

Table 4. Chatbot Accuracy and Performance Metrics

Model Name	Role	Model Type	Latency	Accuracy/Score	Inference Platform
Lead Scoring ML	Intent Classification	Logistic Regression (TF-IDF)	<100ms	97.0% Accuracy (Test Set)	Scikit-learn (CPU)
Mistral 7B	Conversational AI/Q&A	Quantized LLM	2-4 sec/ response	Broad Domain Knowledge	Ollama (Local)
All-Mini LM-L 6-v2	RAG Embeddings	Transformer Encoder	<50ms/ chunk	0.82 (MTEB Score)	Sentence Transformers

5.3 Comparative Business Impact

The automated platform was compared with the existing legacy non-interactive website to measure operational performance and effectiveness during a 30-day simulated implementation period. The combination of the conversational agent and real-time Salesforce synchronization resulted in a 42% decrease in response time initially (from an average of 14 hours to less than 3 minutes). Moreover, the sales team on board experienced a steep 28 percent sales conversion because the representatives were only given pre-qualified leads that were labelled as Hot using the hybrid scoring model.

5.4 Limitations of the Study

In spite of the good performance, this study has various limitations. First, the accuracy of the hybrid model is strongly sensitive to the quality and completeness of the historical CRM data; there is a cold-start problem of new product lines with no historical engagement measures. Second, the RAG chatbot reduces the number of hallucinations; however, it does not support many languages, which restricts its use across international B2B environments. Lastly, the cross-validation has been conducted using a fairly homogeneous dataset of one enterprise provider; i.e., further external validation of the application of the model to highly volatile or vastly different B2B industries is necessary.

6 Conclusion And Future Scope

6.1 Conclusion

This research provided a cohesive framework of an Agentic AI that incorporates the hybrid machine learning lead qualification with NLP-generated engagement directly into the Salesforce CRM framework. The gap between conversational AI and predictive scoring is a resolution of the inefficiencies associated with siloed workflows in B2B sales, in the proposed architecture. The experimental validation asserted that the hybrid models of domain-specific rules and XGBoost algorithms perform better in terms of classification accuracy (97.0%) than conventional baselines. Additionally, it was proven that by deploying an autonomous agent with the ability to operate in an RAG-enabled environment, initial prospect engagement and CRM synchronization can be incorporated effectively, and the human sales effort can be repurposed into valuable strategic negotiations, not through administrative triage.

6.2 Future Scope

Future studies should aim at removing the limitations of the framework. To address the cold-start issue of new B2B markets, the next generation that uses the scoring model should consider few-shot learning and cross-domain transfer learning. Also, the enhancement of the conversational agent to allow real-time and multi-lingual intent recognition will enable the system to be used globally in a much larger scope. Lastly, a more holistic, more financially-oriented criterion of qualification would be the incorporation of predictive deal-value estimation, i.e., not only to predict the probability of conversion, but long-term financial lifetime value of the lead.

References

1. S. Krishna & D. Veeravalli, "The Evolution of CRM: How Salesforce Agentic AI Simplifies Predictive Analytics," International Research Journal of Modernization in Engineering Technology and Science, vol. 10, no. 3, pp. 255–270, 2025.
2. N. Singh & D. Adhikari, "AI-Driven Social Media Analytics for eCommerce Advertising," International Research Journal of Engineering and Technology, vol. 10, no. 12, pp. 509– 515, 2025.
3. A. Barbosa & A. Godoy, "Augmenting Customer Support with an NLP-based Receptionist," arXiv preprint arXiv:2112.01959, 2025.
4. H. Josephine V & Moorthy, "Optimizing Lead Qualification through Machine Learning: A Customer Data-Driven Approach," Advances in Management, vol. 10, no. 3, pp. 255–270, 2024.
5. D. Zumstein, "Marketing Automation & AI Report 2024," ZHAW School of Management and Law, 2024.
6. Salesforce, "Agent-Powered Analytics Deliver Data and Insights in the Flow of Work," Salesforce News, 2024.
7. N. Singh & D. Adhikari, "AI-Powered Social Media Analytics for eCommerce Advertising," International Research Journal of Engineering and Technology, vol. 8, no. 11, pp. 421–428, 2023.
8. ZHAW School of Management and Law, "Marketing Automation and AI Lead Generation Study," Technical Report, 2023
9. A. Barbosa & A. Godoy, "NLP-Based Virtual Receptionist for Customer Service Automation," Proceedings of the International Conference on Natural Language Processing, pp. 89–97, 2021.
10. A. D. Tran, J. Pallant & L. W. Johnson, "Exploring the Impact of Chatbots on Consumer Sentiment and Purchase Behavior in Retail Environments," Journal of Retailing and Consumer Services, vol. 63, article 102718, 2021.
11. T. B. Brown et al., "Language Models are Few-Shot Learners," Advances in Neural Information Processing Systems, vol. 33, pp. 1877–1901, 2020. [Online]. Available: arXiv:2005.14165
12. I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA: MIT Press, 2016.
13. F. Provost and T. Fawcett, "Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking," O'Reilly Media, 2013.
14. S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," Advances in Neural Information Processing Systems, vol. 30, 2017.
15. G. Marcus and E. Davis, "Rebooting AI: Building Artificial Intelligence We Can Trust," Pantheon Books, 2019.
16. Gartner Inc., "Magic Quadrant for Sales Force Automation Platforms," Gartner Research, 2024.
17. Forrester Research, "The State of Conversational AI in Customer Service," Forrester Wave Report, 2024.
18. McKinsey & Company, "The State of AI in 2024: Generative AI's Breakout Year," McKinsey Global Institute, 2024.
19. D. Acharya, K. Kuppan & D. B. Ashwin, "Agentic AI: Autonomous Intelligence for Complex Goals – A Comprehensive Survey," IEEE Access, pp. 1–1, 2025.
20. V. Kotagi, "AI-Augmented CRM: Unifying Data Clouds, Predictive Analytics, and Threat Detection in Salesforce Ecosystems," Recent Trends in Information Technology and Its Application, vol. 9, no. 1, pp. 26–34, 2026.

Acknowledgements

We take this opportunity to express our sincere thanks to our guide and Head of Department, **Dr. Jaya Gupta**, for providing the technical guidelines and suggestions regarding the line of work. We want to express our gratitude for her constant encouragement, support and guidance throughout the development of the project.

We thank our mentor, **Prof. Viki Patil**, for being encouraging throughout the course and for her guidance.

We express our gratitude towards the BE Project Coordinator, **Prof. Sayali Badhan**, for her encouragement during the progress meetings and for providing guidelines to write this report.

We also thank the entire staff of APSIT for their invaluable help rendered during the course of this work. We wish to express our deep gratitude towards all our colleagues of APSIT for their encouragement.

Durvesh Kanade
(22106013)

Vaishnavi Dumbre
(22106115)

Ritik Pandey
(22106054)

Aaryaman Kattali
(22106077)